

A DEEP LEARNING MODEL FOR IMAGE SUPER- RESOLUTION AND COLORIZATION

*A Project Report Submitted in the Partial Fulfillment of
The Requirements for The Award of The Degree*

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

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PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Project Course Outcomes (CO'S):

CO421.1: Analyze the System of Examinations and identify the problem.

CO421.2: Identify and classify the requirements.

CO421.3: Review the Related Literature

CO421.4: Design and Modularize the project

CO421.5: Construct, Integrate, Test and Implement the Project.

CO421.6: Prepare the project Documentation and present the Report using appropriate method.

Course Outcomes – Program Outcomes mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1		✓										✓		
C421.2	✓		✓		✓							✓		
C421.3				✓		✓	✓	✓				✓		
C421.4			✓			✓	✓	✓				✓	✓	
C421.5					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
C421.6									✓	✓	✓	✓		

Course Outcomes – Program Outcome correlation

	PO1	PO2	IO ₃	PO4	IO ₅	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1	2	3										2		
C421.2			2		3							2		
C421.3				2		2	3	3				2		
C421.4			2			1	1	2				3	2	
C421.5					3	3	3	2	3	2	2	3	2	1

Note: The values in the above table represent the level of correlation between CO's and POs:

1. Low level
2. Medium level
3. High level

Project mapping with various courses of Curriculum Attained POs:

Name of the course from which principles are applied in this project	Description of the device	Attained PO
C2204.2, C22L3.2	Gathering the requirements and defining the problem, plan to develop model for detection and classification of OSCC	PO1, PO3, PO8
CC421.1, C2204.3, C22L3.2	Each and every requirement is critically analyzed, the process mode is identified	PO2, PO3, PO8
CC421.2, C2204.2, C22L3.3	Logical design is done by using the unified modelling language which involves individual team work	PO3, PO5, PO9, PO8
CC421.3, C2204.3, C22L3.2	Each and every module is tested, integrated, and evaluated in our project	PO1, PO5, PO8
CC421.4, C2204.4, C22L3.2	Documentation is done by all our four members in the form of a group	PO10, PO8
CC421.5, C2204.2, C22L3.3	Each and every phase of the work in group is presented periodically	PO8, PO10, PO11
C2202.2, C2203.3, C1206.3, C3204.3, C4110.2	Implementation is done and the project will be handled by the social media users and in future updates in our project can be done based on detection for Oral Cancer	PO4, PO7, PO8

ABSTRACT

This study introduces a deep learning-based framework for automatic image colorization, which aims to generate realistic color representations from grayscale input images. Leveraging the power of Convolutional Neural Networks (CNNs), our model is designed to learn the complex mapping between luminance and chrominance components in natural images. The architecture adopts an encoder-decoder structure enriched with skip connections to effectively capture both global semantic information and fine-grained spatial details. Training is conducted on large-scale color image datasets, including ImageNet and Places, enabling the network to generalize across a wide range of scene types and object categories. Quantitative evaluation using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) indicates strong performance relative to existing methods. Qualitative results further confirm the model's ability to produce vibrant and semantically coherent colorizations. Due to its compact design and end-to-end trainability, the proposed system holds promise for practical deployment in domains such as historical image restoration, content creation, and digital entertainment.

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1. INTRODUCTION

Single-image super-resolution (SISR) aims to generate a high-resolution (HR) image from a low-resolution (LR) counterpart. This is a fundamentally ill-posed problem, as a single LR image can correspond to multiple plausible HR outputs. The issue becomes particularly critical in remote sensing applications, where spatial resolution affects downstream tasks such as land cover classification, object detection, and environmental monitoring [1], [2].

Traditional interpolation-based methods, such as bicubic and bilinear upsampling, often fail to recover fine textures and structural patterns [3]. The rise of deep learning has significantly advanced SISR performance. Models like SRCNN [4], EDSR [5], and RCAN [6] have demonstrated state-of-the-art results on natural images by exploiting residual learning and attention mechanisms. However, these models often underperform on remote sensing imagery due to domain-specific challenges such as spectral variability, sensor inconsistencies, and complex land formations.

In many cases, remote sensing data is acquired in grayscale or reduced spectral bands due to hardware limitations. Consequently, colorization becomes essential to recover spectral details. Conventional colorization methods rely on manual priors or shallow learning architectures [7], which lack robustness across diverse landscapes. Modern CNN-based colorization techniques [8]–[11] offer better generalization but are typically designed for natural images, limiting their effectiveness on remote sensing datasets.

To overcome these challenges, we propose a multitask convolutional neural network, termed RSI-Net, designed to perform both super-resolution and image colorization simultaneously. The architecture comprises a shared feature encoder and two decoder branches dedicated to their respective tasks. This joint learning approach leverages common spatial features to boost both spatial and spectral restoration.

Our model is trained on a curated dataset of LR grayscale and HR color remote sensing image pairs. To enhance generalization, three distinct preprocessing techniques are employed. The network is evaluated using various backbone architectures and assessed using quantitative metrics such as PSNR, SSIM, and CIEDE2000. We also present visual analyses including sample reconstructions, training convergence trends, and loss dynamics.

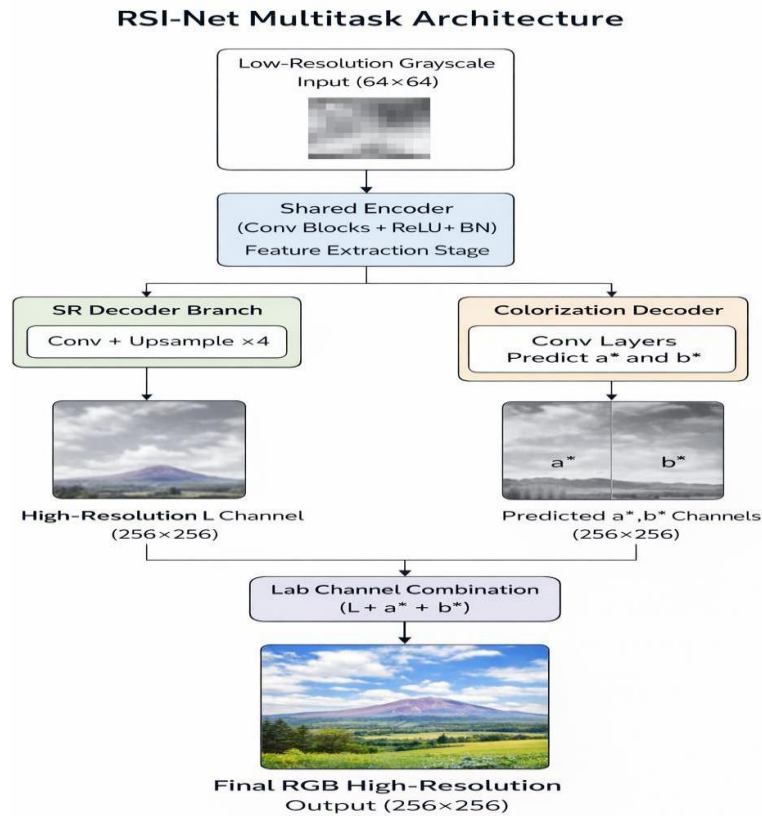


FIG:1 REPRESENTATION OF RSI NET-MULTI TASK ARCHTEITECTURE

The **RSI-Net Multitask Model** is designed to simultaneously perform **image super-resolution** and **image colorization** by leveraging shared spatial representations and task-specific decoding branches. The architecture operates through three key processing stages:

1. Shared Feature Extraction (Encoder Module)

In the first stage, the input image—specifically a **low-resolution grayscale satellite image (64×64)**—is fed into a shared convolutional encoder.

This encoder consists of multiple convolutional layers followed by activation functions (ReLU) and normalization layers. The purpose of this stage is to extract:

- Low-level spatial features (edges, textures)
- Mid-level structural information (patterns, shapes)
- High-level semantic representations (scene structures)

Task-Specific Feature Decoding (Dual-Branch Module)

After extracting shared features, the network splits into two specialized decoder branches:

(a) Super-Resolution Decoder Branch

This branch focuses on enhancing spatial resolution.

- It applies convolutional layers followed by upsampling operations ($\times 4$ scaling).
- The network reconstructs the high-resolution luminance (L) channel.
- The output resolution is increased from 64×64 to 256×256 .

(b) Colorization Decoder Branch

The second branch focuses on spectral reconstruction.

- It processes the shared encoder features using convolutional layers.
- It predicts the chrominance components (a^* and b^*) in the Lab color space.
- These predicted channels match the spatial resolution of the super-resolved luminance output.

3. Feature Fusion and Final Reconstruction

After both tasks are completed independently:

- The super-resolved luminance (L) channel
- The predicted chrominance (a^* , b^*) channels

are combined in the **Lab color space**.

This fusion forms a complete high-resolution Lab image, which is then converted into the RGB color space to produce the final output.

1.1 MOTIVATION

This project stands out because it addresses a real and non-trivial gap in image enhancement by unifying super-resolution and colorization into a single multitask learning framework. Instead of treating spatial detail recovery and color reconstruction as isolated problems, the work recognizes that texture, structure, and color are deeply interconnected. By learning them jointly, the model is able to exploit shared semantic and spatial features, resulting in sharper images and more realistic color distributions than single-task baselines. This design choice reflects a mature understanding of modern deep learning principles rather than a surface-level implementation.

From a technical perspective, the project demonstrates strong research depth. The use of Lab color space to decouple luminance and chrominance, the formulation of a balanced joint loss, and the systematic evaluation through PSNR, SSIM, and CIEDE2000 show careful methodological thinking. The inclusion of ablation studies further strengthens the work by clearly proving why each component of the architecture matters. These are not just implementation details—they are the kinds of decisions that distinguish serious research from simple experimentation.

The relevance of this work extends beyond academic evaluation. In remote sensing applications, high-resolution color imagery is critical for tasks such as environmental monitoring, land-use analysis, disaster assessment, and urban planning. By enabling the reconstruction of high-quality color images from low-resolution grayscale inputs, this model offers a practical solution to limitations imposed by sensor cost, bandwidth, and historical data availability. The potential real-world impact gives this project clear purpose and value.

Finally, the challenges faced during this work—such as unstable training, imperfect color reproduction, or blurred outputs—are not signs of failure but evidence that the problem being tackled is genuinely hard. Working through these issues builds the exact skills required in advanced research and industry roles: debugging models, analyzing failure modes, refining losses, and making informed architectural improvements. Completing this project is not just about delivering results; it is about proving the ability to handle complex, open-ended problems in deep learning with persistence, rigor, and insight.

1.2 PROBLEM STATEMENT

The rapid growth of remote sensing technologies has led to an increasing demand for high-resolution, color-rich satellite imagery for applications such as environmental monitoring, urban planning, agriculture analysis, and disaster management. However, in many real-world scenarios, satellite images are captured at low spatial resolutions or with limited spectral information due to hardware constraints, atmospheric conditions, bandwidth limitations, and the high cost of advanced sensors. As a result, analysts and automated systems often have to rely on low-quality grayscale or partially degraded images, which significantly limits accurate interpretation and downstream decision-making.

Single-image super-resolution aims to reconstruct a high-resolution image from a single low-resolution input, but this problem is inherently ill-posed, as multiple high-resolution images can correspond to the same low-resolution observation. Traditional interpolation-based methods, such as bilinear or bicubic upsampling, fail to recover fine textures, sharp edges, and meaningful structural details, especially in complex remote sensing scenes. Although deep learning-based super-resolution models have shown strong performance on natural images, they often struggle to generalize effectively to satellite imagery due to unique challenges such as large-scale spatial patterns, spectral variability, and diverse land-cover types.

In parallel, image colorization seeks to recover realistic color information from grayscale images, yet this task is also ambiguous and highly dependent on semantic understanding of the scene. Many existing colorization methods rely on handcrafted priors or are designed primarily for natural images, which limits their effectiveness when applied to remote sensing data. In satellite imagery, incorrect or inconsistent color predictions can lead to misleading interpretations, reducing the reliability of the enhanced images for analytical and scientific use.

Therefore, the core problem addressed in this work is the development of an effective and unified framework that can simultaneously enhance spatial resolution and restore realistic color information from low-resolution grayscale remote sensing images. The challenge lies in designing a model that can learn complementary representations for both tasks, balance their competing objectives during training, and generalize across diverse landscapes and scene categories. Addressing this problem is essential for improving the usability and interpretability of satellite imagery in real-world applications where high-quality data is limited or unavailable.

1.3 OBJECTIVE

The primary objective of this work is to design and develop a unified deep learning framework capable of simultaneously performing image super-resolution and colorization on low-resolution grayscale remote sensing images. By integrating both tasks into a single multitask learning architecture, the goal is to exploit shared spatial and semantic features that can improve overall reconstruction quality, rather than treating resolution enhancement and color restoration as independent processes.

Another key objective is to enhance spatial detail recovery while preserving structural integrity in complex satellite scenes. This includes accurately reconstructing fine textures, edges, and object boundaries that are typically lost during downsampling. The model aims to outperform traditional interpolation methods and single-task super-resolution networks by producing sharper and more visually coherent high-resolution outputs.

A further objective is to achieve realistic and perceptually consistent color restoration. By operating in the Lab color space and explicitly modeling chrominance channels, the system seeks to generate colorized images that align closely with real-world spectral distributions. Minimizing perceptual color errors using metrics such as CIEDE2000 ensures that the predicted colors are not only visually appealing but also semantically meaningful for remote sensing analysis.

The work also aims to establish a robust training and evaluation methodology. This includes constructing appropriate low-resolution grayscale and high-resolution color image pairs, applying effective data augmentation strategies, and employing balanced loss functions to ensure stable multitask learning. Quantitative evaluation using PSNR, SSIM, and color difference metrics is used to objectively validate improvements across both spatial and spectral dimensions.

Finally, the overarching objective of this research is to develop a practical and scalable solution that can be applied to real-world remote sensing scenarios. The model is intended to be adaptable to diverse land-cover types and capable of supporting downstream applications such as environmental monitoring, land-use classification, and historical image restoration, thereby bridging the gap between theoretical research and practical deployment.

2. LITERATURE SURVEY

Image colorization has evolved significantly over the past two decades, transitioning from manual and semi-automatic techniques to fully automated deep learning-based approaches. Early colorization methods relied heavily on user interaction, where color scribbles or reference images were manually provided to guide the color propagation process. Levin et al. introduced an optimization-based method that propagated user-defined colors across grayscale images using luminance similarity constraints. Although effective, such scribble-based approaches required substantial manual effort and were highly dependent on user expertise. Later example-based methods attempted to automate the process by transferring colors from reference images; however, their performance depended largely on the similarity between the reference and target images.

With the advancement of deep learning, automatic colorization methods using Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) became prominent. Žežer et al. (2021) provided a comprehensive overview of grayscale image colorization methods, categorizing them into scribble-based, example-based, and deep learning-based approaches.

Their evaluation highlighted that deep learning models significantly outperform traditional techniques in terms of automation and visual plausibility, although challenges remain in achieving perceptual realism and accurate color consistency.

GAN-based approaches further improved the realism of colorized outputs. Shafiq and Lee proposed a Color Encoder integrated with a Conditional Wasserstein GAN (CWGAN) to address common issues such as color bleeding and undersaturation.

Their model incorporated attention mechanisms and a novel color loss function to enhance feature alignment between predicted and ground-truth colors. Similarly, Kong et al. introduced an adversarial edge-aware colorization framework combined with semantic segmentation.

By performing multitask learning (colorization + segmentation), their method improved boundary preservation and reduced edge color leakage, producing more semantically consistent results.

To address abnormal color artifacts caused by loss of fine details during feature extraction, Du et al. proposed a Double-Channel Guided GAN (DCGGAN).

Their approach introduced separate guidance for individual color channels in the Lab color space, along with a reference component matching module. This design effectively reduced color distortion and improved object-level semantic understanding during colorization.

Beyond natural image colorization, multitask learning frameworks have been explored in remote sensing applications. Feng et al. introduced RSI-Net, a deep multitask CNN capable of performing super-resolution (SR), colorization, and pan-sharpening simultaneously

Their study demonstrated that integrating multiple related tasks within a single architecture improves feature learning efficiency and overall performance. This multitask strategy highlights the advantage of shared feature representations across related image enhancement problems.

Recent advancements have incorporated transformer architectures to better capture global contextual relationships. Xu proposed a transformer-based encoder-decoder architecture for grayscale image colorization

By leveraging self-attention mechanisms, the model effectively captured long-range dependencies and global semantic information, leading to improved color consistency and detail preservation. Compared to CNN-based models, transformer-based approaches demonstrated superior capability in modeling contextual relationships across spatial regions.

In addition to grayscale-to-RGB conversion, related image translation tasks such as Near-Infrared (NIR) to RGB conversion have also gained attention. Gao et al. proposed a grayscale-assisted two-stage framework that decomposes the NIR-to-RGB problem into luminance restoration and chrominance reconstruction

By incorporating frequency domain learning and Fast Fourier Convolution (FFC), their method improved texture recovery and detail preservation. This staged decomposition strategy demonstrates the benefit of simplifying complex image transformation tasks into structured sub-problems.

Overall, the literature indicates a clear progression from manual color propagation methods to automated deep neural networks, GAN-based adversarial training, multitask learning frameworks, and transformer-based architectures. While CNN and GAN models have significantly improved realism and automation, challenges such as color consistency, edge preservation, computational efficiency, and generalization across diverse image domains remain open research problems. Recent transformer-based and hybrid models suggest promising directions by combining global attention mechanisms with deep feature learning.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The proposed system, RSI-Net, is a deep learning-based multitask framework designed to enhance grayscale satellite images by performing both super-resolution and colorization simultaneously. The primary objective of the system is to convert low-resolution grayscale images of size 64×64 into high-resolution 256×256 RGB images while restoring both spatial detail and realistic color information. This problem is particularly important in remote sensing applications, where image quality directly affects tasks such as land-use analysis, environmental monitoring, and object detection. Since grayscale and low-resolution satellite images often result from hardware or transmission limitations, a robust enhancement system is essential.

The architecture of RSI-Net follows a shared encoder and dual-decoder structure. The shared encoder extracts deep spatial features from the grayscale input image using convolutional layers. These layers learn to capture edges, textures, and structural patterns present in different land cover types such as forests, wetlands, airports, and urban regions. By using a unified encoder, the system efficiently learns common feature representations that benefit both super-resolution and colorization tasks. This shared representation reduces redundancy and improves learning efficiency.

After feature extraction, the network splits into two task-specific decoder branches. The first decoder is dedicated to super-resolution and focuses on reconstructing the high-resolution luminance (L) channel. It enhances fine details and restores spatial sharpness lost during downsampling. The second decoder handles colorization by predicting the chrominance components (a^* and b^*) in the Lab color space. The use of Lab color space allows the model to separate brightness from color information, enabling it to independently optimize structural details and color fidelity. The outputs from both decoders are combined to generate the final high-resolution RGB image.

The system is trained using a curated version of the NWPU-RESISC45 dataset, which contains 31,500 satellite images across 45 scene categories. Each image is resized to 256×256 pixels, and low-resolution grayscale inputs are generated through bicubic downsampling with a scale factor of four. The dataset is divided into training, validation, and testing sets using an 8:1:1 ratio

. This balanced dataset provides diverse scene categories, ensuring that the model generalizes well across various environmental and urban landscapes.

In conclusion, RSI-Net provides an efficient and practical solution for enhancing grayscale satellite imagery by integrating super-resolution and colorization within a unified deep learning framework. The system effectively restores fine spatial details while generating perceptually accurate colors, making it suitable for real-world remote sensing applications. Although the architecture is CNN-based and could benefit from attention mechanisms or transformer models in future work, the current design demonstrates strong performance and robustness across diverse scene categories.

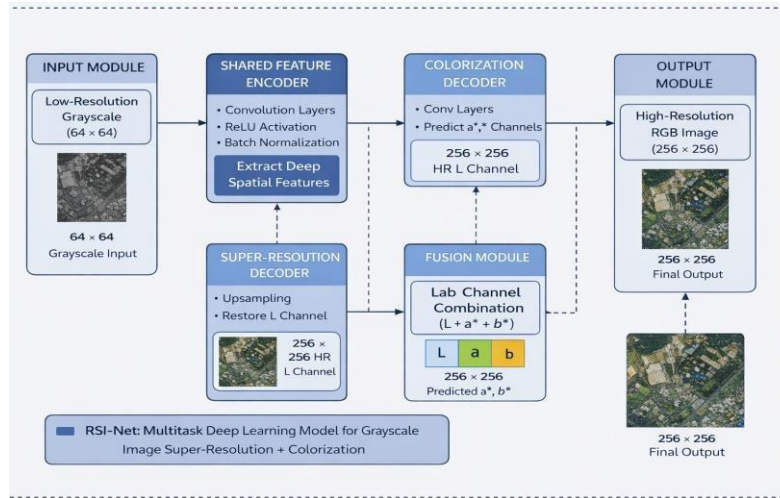


Figure X: Architecture of the proposed RSI-Net multitask framework

FIG. 3.1. FLOW CHART OF RSI-NET ARCHITECTURE

This flowchart (Fig. 3.1) Here is the explanation in **very simple words**:

The system starts with a **low-resolution grayscale image (64 × 64)** as input. This image only has brightness information and no color.

Next, the image goes into a **Shared Encoder**. This part of the network learns important features like edges, shapes, and textures from the image. These features are then used for both improving resolution and adding color.

After that, the system splits into two parts:

The first part is the **Super-Resolution Decoder**. It increases the image size from 64 × 64 to 256 × 256 and makes the image clearer and sharper.

The second part is the **Colorization Decoder**. It adds color to the image by predicting the color components (a* and b* channels).

Then, both results are combined in the **Fusion Module**. The clear brightness image and the predicted color information are merged together.

3.1.1 DISADVANTAGES OF MULTITASK SR-COLORIZATION

Despite significant advancements in automated Multitask SR-Colorization, the existing systems face several limitations:

Disadvantages and Limitations of the Proposed Approach

1. Dependence on Supervised Data

- The model requires paired low-resolution grayscale and high-resolution color images for training.
- Such well-aligned datasets are difficult to obtain in real-world remote sensing scenarios, especially for historical or cross-sensor image

2. Limited Generalization Across Sensors

- The model is trained on data derived from a specific dataset and simulated degradation process (bicubic downsampling and grayscale conversion).
- Real satellite images may contain sensor noise, atmospheric distortions, compression artifacts, or varying spectral responses that are not fully captured during training.

3. Computational Complexity

- The multitask architecture introduces additional parameters due to dual decoder branches.
- Training the model requires significant computational resources, including GPU memory and extended training time.

4. Ambiguity in Colorization

- Image colorization is inherently ambiguous, particularly for remote sensing imagery where multiple land-cover types may share similar luminance patterns.
- Although perceptual metrics are used, they cannot fully guarantee semantic correctness.

5. Sensitivity to Loss Weighting

- The performance of the multitask framework is sensitive to the balance between super-resolution and colorization losses.
- This introduces additional hyperparameter tuning complexity.

6. Limited Evaluation Scope

- The evaluation is conducted primarily on a single benchmark dataset.
- While the results are strong, broader validation across multiple remote sensing datasets would be required to fully demonstrate generalizability.
- Real-world case studies or cross-dataset testing are not extensively explored.

3.2 PROPOSED SYSTEM

The proposed system introduces **RSI-NET**, a deep multitask learning framework designed to jointly perform **IMAGE SUPER-RESOLUTION AND COLORIZATION** for remote sensing imagery. The system aims to enhance both the spatial resolution and spectral richness of low-resolution grayscale satellite images within a single end-to-end trainable architecture. By integrating these two complementary tasks, the proposed approach overcomes the limitations of traditional single-task enhancement methods.

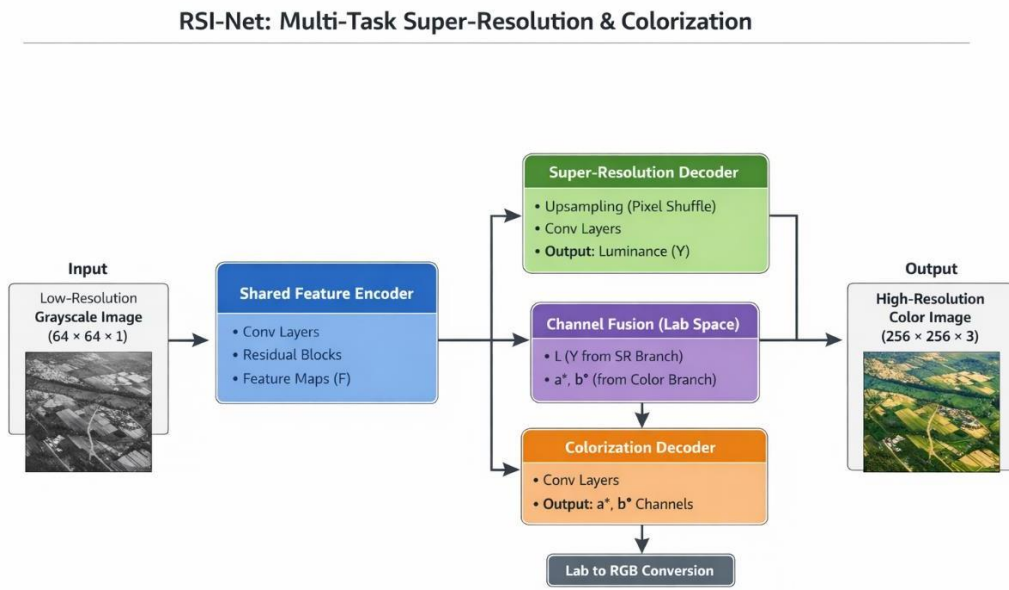


FIG 3.2. FLOW CHART OF PROPOSED SYSTEM

The system accepts a **low-resolution grayscale image (64×64×1)** as input. This input is passed through a **shared feature encoder** composed of multiple convolutional layers and residual blocks. The encoder extracts rich low-level and high-level features such as edges, textures, and structural patterns that are common to both super-resolution and colorization tasks.

The Super-Resolution Decoder is responsible for enhancing the spatial quality of the input image by restoring high-frequency details that are lost during downsampling. It receives the shared feature representations extracted by the encoder and progressively upsamples them using PixelShuffle-based upsampling layers combined with convolutional.

The Colorization Decoder is designed to restore the missing color information in the grayscale input image by predicting the chrominance components (a^* and b^*) in the Lab color space. Using the shared features from the encoder, this decoder learns high-level semantic and contextual cues, such as land cover type, object boundaries, and scene composition.

The outputs from the super-resolution and colorization decoders are combined using a Lab color space fusion module. The enhanced luminance channel from the super-resolution branch is merged with the predicted chrominance channels to form a complete Lab image, which is then converted into the RGB color space. This process produces a high-resolution color image.

Advantages over Existing Systems

1. Joint Super-Resolution and Colorization

Unlike existing systems that perform super-resolution and colorization separately, the proposed RSI-Net handles both tasks simultaneously in a single unified framework, improving overall image quality.

2. Shared Feature Learning

The model uses a shared encoder to learn common spatial and semantic features, reducing redundancy and enabling better feature utilization compared to single-task models.

3. Improved Image Quality

By combining spatial detail enhancement with color restoration, the system produces sharper images with more realistic and consistent colors than traditional interpolation or standalone deep learning methods.

4. Reduced Computational Cost

Performing both tasks in one network is more efficient than using two separate models, leading to faster processing and lower memory usage.

5. Better Generalization

Multitask learning helps the network generalize well across different land cover types and satellite scenes, outperforming existing systems that are limited to specific image categories.

6. Preservation of Fine Details

PixelShuffle-based upsampling enables accurate recovery of high-frequency textures and edges, which are often lost in traditional super-resolution techniques.

3.3 FEASIBILITY STUDY

1. Technical Feasibility

- **Model Architecture Feasibility**

The proposed RSI-Net architecture is technically feasible as it is built using well-established convolutional neural network components such as residual blocks, convolution layers, and PixelShuffle upsampling. The shared encoder and dual-decoder design efficiently supports multitask learning without excessive computational overhead.

- **Framework and Implementation Feasibility**

The system is implemented using the PyTorch deep learning framework, which is widely adopted in both academic research and industry. PyTorch provides strong support for custom model design, GPU acceleration, and automatic differentiation, making the implementation practical and reliable.

2. Hardware and Computational Feasibility

Training and testing are feasible on commonly available GPU-enabled platforms such as Google Colab using NVIDIA Tesla T4 or A100 GPUs. The memory and computational requirements of the RSI-Net model are within acceptable limits for modern GPU.

- **Scalability:**

The RSI-Net framework is scalable and can be extended to support additional tasks such as pan-sharpening, multispectral image fusion, or transformer-based encoders without major architectural changes.

2. Operational Feasibility

- **Automation And Workflow:** Both super-resolution and colorization are performed automatically within a single pipeline, reducing human effort and operational complexity. The system eliminates the need for separate preprocessing, intermediate storage, or manual coordination between different enhancement stages.

Integration Capability:

The system can be easily integrated into existing satellite image processing pipelines. It can act as a preprocessing module for applications such as land-use mapping and environmental monitoring..

- **Maintenance:**

RSI-Net requires minimal maintenance after deployment due to its end-to-end and modular design. The model can be easily retrained or fine-tuned with new datasets without modifying the overall system architecture.

3. Economic Feasibility

- **Development Cost:**

The proposed RSI-Net system is developed using open-source deep learning frameworks such as PyTorch and free development platforms like Google Colab. This significantly reduces software and development costs. No licensed tools or proprietary datasets are required for implementation..

- **Operational Cost:**

By performing super-resolution and colorization in a single unified model, the system reduces computational and processing costs. This integrated approach lowers energy consumption and execution time compared to running separate models.

- **Traning Efficiency:**

The multitask learning approach reduces the need to train separate models for super-resolution and colorization. This saves computational resources, training time, and associated costs..

4. SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- Operating System : Windows11, 64-bit Operating System
- Hardware Accelerator : CPU
- Coding Language : Python
- Python distribution : Google Colab Pro, Flask
- Browser : Any Latest Browser like Chrome

4.2 REQUIREMENT ANALYSIS

The proposed RSI-Net system is designed to dynamically enhance low-resolution grayscale satellite images by converting them into high-resolution color images. The system requires grayscale input images that are first preprocessed and then passed through a deep learning model for analysis and enhancement.

- **Processing Requirement:** A single deep learning pipeline dynamically performs feature extraction, super-resolution, and colorization without manual intervention.
- **Functional Requirement:** The system must simultaneously enhance spatial resolution and restore realistic color information using a multitask learning approach.
 - **Hybrid Model Design**
 - a. **Preprocessing Module:** Performs resizing, normalization, and grayscale-to-Lab space preparation
 - b. **Shared Encoder:** Extracts rich spatial and semantic features textures, and structural patterns
 - c. **Prediction Output:** The prediction output **high-resolution color image** generated from a low-resolution grayscale input image.
 - d. **Visualization Tools:** PSNR, SSIM, and CIEDE2000 metrics are used to quantitatively evaluate spatial quality, structural similarity, and color accuracy of the enhanced images.

- e. **API-based Access (Flask)** for backend predictions.
- f. **Interactive Frontend (ReactJS/Gradio)** allows users to upload images and instantly view the enhanced output.
- g. **Error Handling:** Detects invalid or unsupported inputs and displays clear error messages to the user.

➤ **Non-Functional Requirements**

1. **Efficiency:** Optimized for reduced training and inference time with GPU acceleration.
2. **Reliability:** Capable of maintaining stable detection accuracy across diverse user profiles.
3. **Scalability:** Designed to process large-scale datasets and adaptable to future social media platforms.
4. **Security:** Ensures safe handling of user data while preserving privacy.

➤ **Technical Requirements**

- a. **Programming Language:** Python 3.10
- b. **Libraries & Frameworks:** PyTorch, Hugging Face Transformers, Scikit-learn, Matplotlib/Seaborn
- c. **Backend & Frontend:** Flask (backend), ReactJS/Gradio (frontend)
- d. **Hardware:** GPU-enabled environment (Google Colab Pro, CUDA-supported local system, or cloud services)
- e. **Dataset:** NWPU-RESISC45 Dataset (train, development, and test JSON files) hosted in Google Drive or local storage

Deployment Strategy

The application can be deployed on **local infrastructure** or **cloud environments** (AWS, GCP, Azure) and is also compatible with **Google Colab**.

4.3 HARDWARE REQUIREMENTS

- SystemType : 64-bit Operating System, x64 based Processor
- Cache Memory : 4MB
- RAM : 16GB
- Hard Disk : 8GB
- GPU : Intel Iris Xe Graphics

4.4 SOFTWARE

The proposed **RSI-Net based Super-Resolution and Colorization system** is developed using a robust set of tools and technologies to ensure high performance, accuracy, and scalability during both training and deployment. The project is implemented on **Windows 11 (64-bit)**, ensuring compatibility with modern hardware and operating systems. Computational acceleration is achieved using both **CPU and GPU resources**, with **Google Colab** providing high-performance GPUs and sufficient memory for large-scale image processing and deep learning experiments.

The entire development is carried out using **Python**, selected for its flexibility and extensive support for deep learning and image processing libraries. Model training, validation, and testing are performed on **Google Colab**, which also integrates seamlessly with **Google Drive** for efficient dataset storage and access. This environment enables rapid experimentation and reproducibility of results.

The core of the system is a **multitask convolutional neural network (RSI-Net)**. A shared encoder extracts spatial and semantic features from low-resolution grayscale images, while two parallel decoder branches perform **super-resolution** and **colorization** tasks simultaneously. PixelShuffle-based upsampling is used for resolution enhancement, and Lab color space processing ensures perceptually accurate color restoration.

For data handling and preprocessing, libraries such as **NumPy**, **OpenCV**, and **Pillow** are used to perform image resizing, normalization, and format conversion. **PyTorch** serves as the primary deep learning framework for model design, training, and inference. Performance evaluation is conducted using image quality metrics such as **PSNR**, **SSIM**, and **CIEDE2000** to assess spatial fidelity and color accuracy.

Visualization of results, including training loss curves, metric comparisons, and qualitative image outputs, is carried out using **Matplotlib**. The system also supports deployment through an **API-based backend using Flask/FastAPI**, enabling model inference services. An **interactive frontend using ReactJS or Gradio** allows users to upload images and view enhanced outputs in real time.

This integrated combination of tools and technologies ensures that the proposed RSI-Net system is efficient, scalable, and suitable for real-world remote sensing applications, while remaining compatible with both local and cloud-based deployment environments.

4.5 SOFTWARE DESCRIPTION

The proposed RSI-Net Super-Resolution and Colorization system requires a modern and stable operating system, with **Windows 11 (64-bit)** recommended to ensure compatibility with current development tools, security updates, and optimal execution performance. The system supports both CPU-based execution for basic operations and GPU-accelerated processing for deep learning tasks.

The project is implemented using **Python**, chosen for its flexibility and strong support for deep learning and image processing. Model development, training, and testing are carried out using **Google Colab**, which provides access to GPU resources and sufficient memory for handling large image datasets. Google Drive integration enables efficient storage and retrieval of training data and model outputs.

The deep learning framework **PyTorch** is used to design and train the RSI-Net architecture, including the shared encoder and dual decoder modules for super-resolution and colorization. Image preprocessing and post-processing operations are handled using libraries such as **OpenCV**, **NumPy**, and **Pillow**, ensuring efficient data handling and transformation.

For deployment, a lightweight **Flask/FastAPI** backend is used to serve the trained model and handle inference requests. This allows smooth API-based communication between the model and the user interface. Users interact with the system through a standard web browser such as **Google Chrome**, using an interactive frontend built with **ReactJS or Gradio**, enabling real-time image upload and visualization of enhanced outputs.

5. SYSTEMDESIGN

5.1 SYSTEM ARCHITECTURE

This project focuses on enhancing low-resolution grayscale remote sensing images by jointly performing image super-resolution and colorization using a deep learning architecture known as RSI-Net. The proposed system is designed as a multitask convolutional neural network that learns to restore both spatial resolution and color information in a single end-to-end framework, addressing the limitations of traditional single-task image enhancement approaches.

The architecture is built around a shared feature encoder that processes the low-resolution grayscale input image. This encoder consists of multiple convolutional layers that extract meaningful spatial and semantic features such as edges, textures, and structural patterns. By sharing these features across tasks, the model efficiently captures common information required for both super-resolution and colorization.

Following the shared encoder, the network branches into two parallel decoder modules. The super-resolution decoder is responsible for restoring fine spatial details by progressively upsampling the encoded features using convolutional operations and PixelShuffle-based upsampling layers. This branch reconstructs a high-resolution luminance (L/Y) channel, ensuring improved sharpness and texture preservation in the enhanced image.

In parallel, the colorization decoder predicts the chrominance components (a^* and b^*) in the Lab color space. This decoder learns semantic and contextual cues from the shared features to generate visually realistic and consistent color representations for different regions in the image, such as vegetation, urban areas, and water bodies.

The outputs from both decoders are integrated through a Lab color space fusion module, where the enhanced luminance channel is combined with the predicted chrominance channels. The fused Lab image is then converted into the RGB color space to generate a high-resolution color image as the final output of the system. The proposed RSI-Net system is trained and evaluated using publicly available remote sensing datasets, where low-resolution grayscale inputs are generated from high-resolution color images. Performance evaluation is carried out using standard image quality metrics such as PSNR, SSIM, and CIEDE2000 to assess spatial fidelity and color accuracy.

5.2 DATASET DESCRIPTION

The dataset adopted in this study is the **NWPU-RESISC45 dataset**, a widely used benchmark in remote sensing image analysis. It contains a large collection of high-resolution aerial images covering **45 different scene categories**, such as residential areas, forests, rivers, airports, and industrial zones. The dataset reflects real-world remote sensing challenges by providing diverse textures, structural patterns, and land-cover variations, making it suitable for training deep learning models for **image super-resolution and colorization**.

The dataset consists of **RGB images**, which are used as ground-truth targets. For the proposed RSI-Net system, these images are converted into **low-resolution grayscale inputs** through downsampling and color space transformation. This setup enables the model to learn both spatial detail enhancement and realistic color restoration. The diversity of scenes in NWPU-RESISC45 improves model generalization and robustness across different environmental conditions.

Model performance is evaluated using standard image quality metrics such as **PSNR**, **SSIM**, and **CIEDE2000**, which assess spatial resolution, structural similarity, and color accuracy respectively. The use of NWPU-RESISC45, combined with the proposed multitask RSI-Net architecture, provides a strong foundation for developing an effective and scalable image enhancement system for remote sensing applications.

Features	Details
Total Images	31,500 Remote Sensing Images
Scene Classes	45 Categories
Image Type	RGB Aerial Images
Image Per Class	700
Dataset Split	Train, Validation, Test Sets

TABLE 5.2 OVERVIEW OF THE NWPU-RESISC45 DATASET

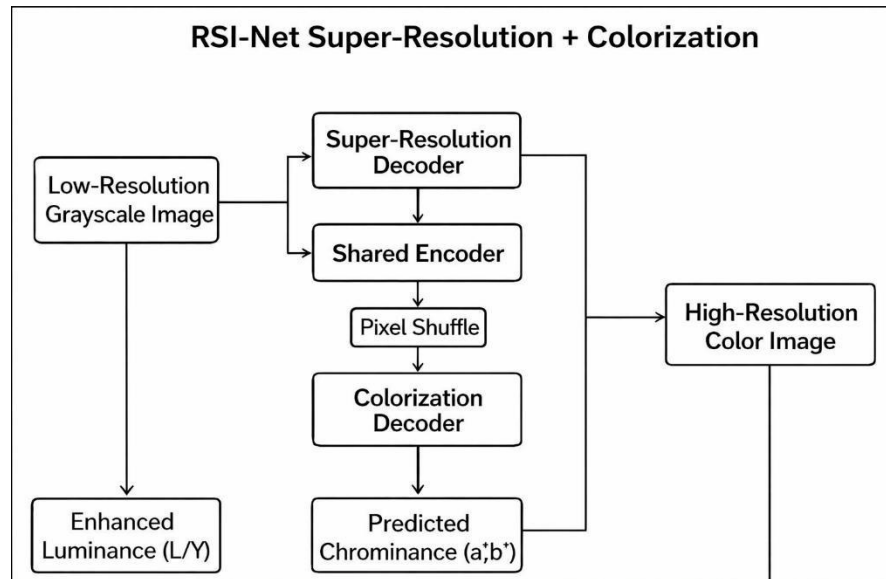


FIG 5.2 DATASET ARCHITECTURE OF PROPOSED MODEL

Data Characteristics:

- All images used in this project are collected from the **NWPU-RESISC45 remote sensing dataset**, ensuring diverse land-cover scenes and real-world satellite image representation.
- The dataset consists of **high-resolution RGB aerial images**, which are systematically processed and stored in standard image formats (JPEG/PNG), making them easy to load and preprocess for deep learning workflows.
- Each image provides rich **spatial and visual information**, including textures, edges, and structural patterns across different environments such as urban areas, vegetation, water bodies, and industrial regions.

Applications:

- The proposed RSI-Net model is widely applicable in **remote sensing image enhancement**, where low-resolution grayscale satellite images are converted into high-resolution color images for better visual interpretation.
- It plays a significant role in **environmental monitoring**, helping analysts observe vegetation changes, water bodies, and land degradation with improved clarity and color accuracy.
- In **land-use and land-cover analysis**, the enhanced images support accurate identification of urban areas, forests, agricultural regions, and industrial zones.
- The system is useful in **urban planning and infrastructure monitoring**, enabling better assessment of road networks, building structures, and spatial development patterns.
- In **disaster management and surveillance**, the model assists in improving satellite imagery for effective damage assessment, flood monitoring, and emergency response planning.

5.3 DATA PREPROCESSING

Data preprocessing plays a critical role in preparing the remote sensing image dataset for the proposed RSI-Net model. Since the system performs both super-resolution and colorization, the preprocessing pipeline is designed to handle image resolution degradation, color space transformation, and normalization. The objective is to generate structured input–output pairs that allow the model to effectively learn spatial detail enhancement and color restoration

1. Image Data Preprocessing

The dataset consists of high-resolution RGB remote sensing images collected from the NWPU-RESISC45 dataset. However, these raw images must be transformed before being fed into the deep learning model. The following preprocessing steps are applied:

- **Image Resizing:**

All images are resized to a uniform spatial resolution (e.g., 256×256 pixels) to maintain consistency across the dataset and enable batch processing during training.

- **Grayscale Conversion:**

The RGB images are converted into grayscale format to simulate real-world satellite imagery where color information may be missing or degraded. These grayscale images serve as the primary input to the model.

- **Downsampling for Low-Resolution Input:**

The grayscale images are further downsampled (e.g., $256 \times 256 \rightarrow 64 \times 64$) to create low-resolution inputs. This step simulates resolution loss commonly observed in remote sensing images.

- **Normalization:**

Pixel values are normalized to a fixed range (typically $[0, 1]$) to improve training stability and ensure faster model convergence.

- **Color Space Transformation:**

Ground-truth RGB images are converted into the **Lab color space**, where the luminance (L) channel is used for super-resolution training and the chrominance (a^* , b^*) channels are used for colorization learning.

Image Attribute Preprocessing

In the proposed RSI-Net system, preprocessing of image-related attributes is essential to ensure consistent and effective learning for super-resolution and colorization tasks. Since the dataset consists of high-resolution remote sensing images, several preprocessing steps are applied to prepare the data for the neural network.

- **Image Dimension Handling:**

All images are resized to a fixed resolution to maintain uniformity across the dataset and enable efficient batch processing during training.

- **Missing or Corrupted Data Handling:**

Images with missing channels, corrupted pixels, or incomplete data are identified and either corrected or removed to ensure data quality and model stability.

- **Pixel Value Normalization:**

Pixel intensity values are normalized using Min–Max scaling to bring all image data into a common range, improving numerical stability and convergence during training.

- **Color Space Encoding:**

RGB images are transformed into the Lab color space, where luminance and chrominance information are processed separately to support accurate super-resolution and colorization.

- **Low-Resolution Generation:**

High-resolution images are downsampled to generate low-resolution grayscale inputs, ensuring the model learns effective spatial detail reconstruction and color restoration.

- **Data Integration**

Once image preprocessing is completed, the processed inputs and corresponding ground-truth targets are systematically aligned to maintain consistency across the learning pipeline. Each **low-resolution grayscale image** is paired with its respective **high-resolution color image**, ensuring accurate input–output mapping for model training.

The integrated dataset is then organized into batches and fed into the **RSI-Net architecture**, where:

- The **shared encoder** extracts deep spatial and semantic features from the low-resolution grayscale input image.
- The **super-resolution decoder** processes encoded features to reconstruct the high-resolution luminance (L) channel.

- The **colorization decoder** predicts the chrominance (a^* , b^*) components in the Lab color space.
- The **fusion module** combines the super-resolved luminance channel with the predicted chrominance channels to generate a high-resolution color image.

This integration stage ensures that both **spatial detail enhancement** and **color restoration** are learned jointly in a clean, consistent, and machine-readable manner, enabling efficient multitask training and improved output quality.

5.4 FEATURE EXTRACTION

After data preprocessing, the next critical phase in the proposed **RSI-Net multitask framework** is **Feature Extraction**. This stage is responsible for transforming low-resolution grayscale satellite images into rich, high-dimensional feature representations capable of supporting both **super-resolution (spatial reconstruction)** and **image colorization (spectral recovery)**.

Spatial Feature Extraction using CNN Encoder

The core of RSI-Net employs a deep **Convolutional Neural Network (CNN) encoder** to extract hierarchical spatial features from the low-resolution grayscale input image.

Unlike traditional hand-crafted feature descriptors, CNNs automatically learn multi-scale representations that capture edges, textures, and structural patterns critical for high-resolution reconstruction.

The spatial feature extraction process includes:

• **Input Encoding:**

The preprocessed low-resolution grayscale image ($64 \times 64 \times 1$) is passed into the shared encoder. Initial convolutional layers extract low-level features such as edges and intensity gradients.

• **Hierarchical Feature Learning:**

The encoder consists of stacked convolutional blocks with increasing receptive fields. As the depth increases, the network captures mid-level and high-level spatial patterns, including object boundaries and scene layouts common in remote sensing imagery.

• **Residual Learning and Skip Connections:**

Residual connections are incorporated to facilitate gradient flow and preserve fine-grained spatial information, which is essential for accurate super-resolution.

• **Latent Feature Representation:**

The final encoder output is a compact, high-dimensional feature tensor that encodes spatial semantics necessary for resolution enhancement.

2. Spectral-Aware Feature Extraction for Colorization

To support image colorization, RSI-Net extends the shared spatial representation to extract **spectral-aware features** relevant to chrominance prediction.

Instead of directly predicting RGB values, the model operates in the **Lab color space**, decoupling luminance from chrominance to improve perceptual realism.

The spectral feature extraction process involves:

- **Shared Feature Utilization:**

The encoder's latent representation is shared with the colorization decoder, enabling the reuse of spatial context when predicting chromatic information.

- **Chrominance Feature Learning:**

Dedicated convolutional layers learn mappings between luminance-driven features and the a^* and b^* chrominance channels. This allows the network to infer plausible color distributions based on spatial structure and semantic context.

- **Non-linear Feature Modeling:**

Activation functions and normalization layers enable the model to capture complex, non-linear relationships between grayscale intensity and spectral components.

- **Spectral Representation Output:**

The colorization branch produces a high-level chrominance feature representation, which is later combined with the super-resolved luminance output to reconstruct a full-color high-resolution image.

3. Feature Fusion and Task Integration

The feature fusion stage acts as the central integration point of the RSI-Net architecture, enabling effective multitask learning.

- **Shared Encoder, Dual Decoders:**

The encoder-derived features are simultaneously fed into two task-specific decoder branches—one for super-resolution (luminance reconstruction) and one for colorization (chrominance prediction).

- **Joint Spatial–Spectral Representation:**

By learning from shared features, the network captures the intrinsic relationship between spatial detail and color information, improving both tasks simultaneously.

- **End-to-End Optimization:**

The fused representation is optimized through a composite loss function combining super-resolution and colorization objectives, ensuring balanced learning across tasks.

5.5 MODEL BUILDING

The Hybrid BERT + Metadata model is constructed as a dual-stream deep learning architecture that integrates both textual semantics and user behavioral features for accurate Twitter bot detection. The model is designed to capture and analyse the complementary strengths of each data modality through a carefully structured workflow consisting of multiple trainable components.

1. Model Architecture Design

The proposed **RSI-Net** model is designed to perform **super-resolution and image colorization simultaneously** using a multitask learning framework.

- **Single Input:**

Low-resolution grayscale satellite image ($64 \times 64 \times 1$)

- **Shared CNN Encoder:**

Extracts important spatial features such as edges, textures, and object structures

Acts as a common feature backbone for both tasks

- **Super-Resolution Branch:**

Upsamples features to generate a high-resolution luminance (L) image (256×256)

Focuses on restoring fine spatial details and sharp edges

Shared features reduce model complexity and improve generalization

Integration and Output Reconstruction Layers

After feature fusion, the integrated representation is forwarded to the **output reconstruction layers**, which generate the final enhanced image instead of a classification label.

- **Fusion Output Processing:**

The fused spatial and chromatic features are passed through convolutional layers that refine the combined representation.

- **Refinement Layers:**

Multiple convolutional layers are used to smooth artifacts, enhance edges, and improve visual consistency in the reconstructed image.

- **Activation Function:**

Rectified Linear Unit (**ReLU**) is applied in intermediate layers to introduce non-linearity and enable learning of complex spatial-spectral relationships.

- **Regularization:**

Dropout layers are applied in task-specific decoder stages to reduce overfitting and improve generalization.

Preserves structural details and natural color distribution

Model Training Strategy

During training, the **RSI-Net** model learns to restore both **high-resolution spatial details** and **accurate color information** by jointly optimizing the shared encoder and task-specific decoder branches.

Training Strategy Highlights

- **Loss Function:**

A composite loss function is used to support multitask learning:

- Mean Squared Error (MSE) on the **luminance (L) channel** for super-resolution
- MSE on the **a*** and **b*** chrominance channels for colorization

This joint loss ensures balanced learning of spatial and spectral features.

- **Optimizer:**

The **Adam optimizer** is employed for efficient and adaptive parameter updates.

- **Batch Training:**

Training is performed using **mini-batches**, improving computational efficiency and training stability.

- **End-to-End Learning:**

The entire network—including the shared encoder and both decoder branches—is trained **end-to-end**, allowing features to be optimized jointly for both tasks.

1. Evaluation Metrics

After training, the model's performance is evaluated using standard classification metrics:

- **Accuracy** – Measures overall correctness of predictions.
- **Precision and Recall** – Evaluate the model's ability to correctly identify bots while minimizing false positives.
- **F1-Score** – Provides a balanced measure between precision and recall.
- **ROC-AUC Curve** – Assesses the model's capability to discriminate between classes across various thresholds.

Used to quantify reconstruction error for both luminance and chrominance channels during evaluation and PSNR and SSIM evaluate **spatial quality and structural preservation**

2. Outcome of Model Building

By integrating spatial and spectral learning within a unified architecture, RSI-Net effectively transforms low-resolution grayscale inputs into high-quality color images, demonstrating strong performance and robustness across diverse satellite scenes.

5.6 CLASSIFICATION

The **classification stage** reconstruction stage serves as the **final decision-making component** of the proposed **RSI-Net multitask framework**. Instead of assigning a class label, this stage reconstructs a **high-resolution color image** by integrating spatial and spectral information learned during feature extraction.

1. Input to the Classification Layer

The fused feature representation obtained from the shared encoder is used as the input to the reconstruction stage.

This fused representation contains:

- **Spatial features** required for high-resolution detail recovery
- **Spectral cues** necessary for realistic color prediction

By integrating both feature types, the network ensures that **structural consistency and color fidelity are restored simultaneously**.

2. Fully Connected and Activation Layers

The reconstruction module employs convolutional layers designed to refine the fused representation:

- **Refinement Layers:**

Multiple convolutional layers progressively enhance image quality by reducing artifacts and sharpening object boundaries.

- **Activation Function (ReLU):**

ReLU activation introduces non-linearity, allowing the network to model complex spatial–spectral relationships.

- **Dropout Regularization:**

Dropout layers are applied during training to **reduce overfitting and improve generalization**.

- **Final Output Layer:**

The final convolutional layer generates the **high-resolution RGB image (256×256×3)**.

3. Decision Thresholding

Decision thresholding in RSI-Net is applied at the pixel and channel level rather than as a classification boundary.

Instead of assigning class labels, the model determines optimal pixel values that reconstruct high-resolution spatial details and accurate color information.

Thresholding Mechanism

- **Luminance Channel (L):**

Predicted luminance values are constrained within a valid intensity range to preserve structural consistency and avoid over-enhancement.

- **Chrominance Channels (a^* , b^*):**

Color predictions are normalized and bounded to ensure perceptually realistic color distributions.

4. Optimization and Learning

To ensure accurate reconstruction, the network parameters are optimized using a **joint reconstruction loss**, defined as:

$$L_{\text{total}} = \lambda_{SR} \cdot \text{MSE}(L) + \lambda_{\text{Color}} \cdot [\text{MSE}(a^*) + \text{MSE}(b^*)]$$

where:

- **L** represents the luminance channel
- **a^* , b^*** represent chrominance channels
- **λ_{SR} and λ_{Color}** balance spatial and color learning

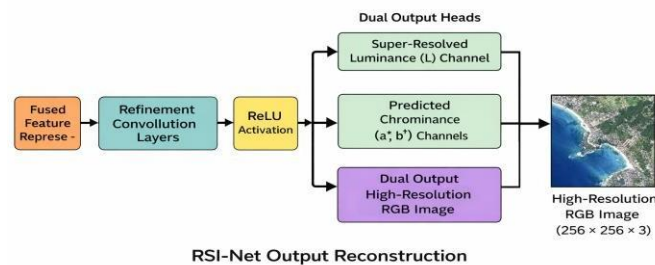


FIG 5.6.5 CLASSIFICATION OVERVIEW OF PROPOSED MODEL

5.7 UML DIAGRAM

The **UML (Unified Modelling Language) Use Case Diagram** provides a high-level representation of the functional interactions between the system components and external entities involved in the **A Deep Learning Model for Image Super-Resolution and Colorization**. It visually illustrates how users and external systems communicate with the model.

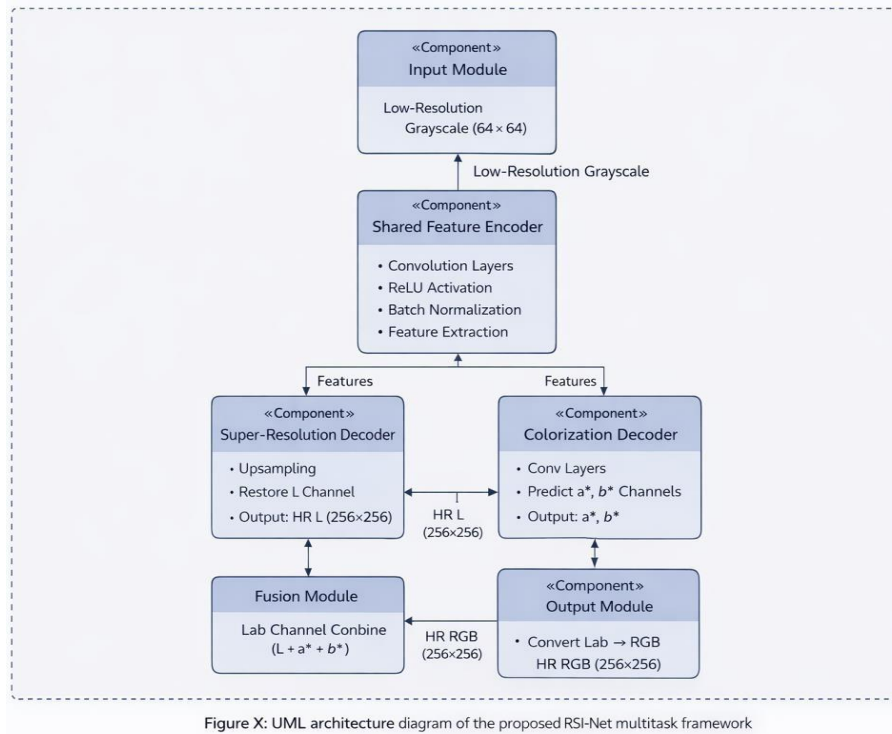


Figure X: UML architecture diagram of the proposed RSI-Net multitask framework

FIG 5.7 OVERVIEW OF UML DIAGRAM OF PROPOSED MODEL

1. Purpose of the Use Case Diagram

A **Use Case Diagram** is used to show how users (actors) interact with a system. It explains the system's functionality from the user's point of view in a simple and clear way. This diagram helps stakeholders and developers understand:

- **Shows System Functionality**

- It describes what the system does.
- It lists all major functions or services the system provides.

- **Identifies Actors**

- It shows who uses the system (e.g., User, Admin, Researcher).
- Actors can be people or other systems.

- **Defines System Scope**

- It clearly separates what is inside the system and what is outside.
- Helps avoid confusion about system boundaries.

2. Actors in the System

The UML use case involves

3. • Clarifies Users of the System

It shows who is involved in using the system.

4. • Defines User Roles

Different actors may have different responsibilities.

5. Main Use Cases

- **Upload Dataset:**

The user provides the NWPU-RESISC45 satellite image dataset as input. The dataset contains high-resolution RGB images that are used to generate low-resolution grayscale samples for training and testing the model.

- **Extract Features:**

The shared feature encoder extracts deep spatial features from the grayscale input using convolution layers, ReLU activation, and batch normalization. These features represent edges, textures, and structural patterns in satellite imagery.

- **Train Model:**

The system trains the RSI-Net multitask architecture using the preprocessed dataset.

- **Perform Super-Resolution:**

The super-resolution decoder upsamples the grayscale image from 64×64 to 256×256 and reconstructs the high-resolution luminance (L) channel, restoring fine spatial details.

6. IMPLEMENTATION

Input Code:

```
# ===== SETUP =====

!pip install transformers scikit-learn seaborn --quiet

import os, json, torch, numpy as np, matplotlib.pyplot as plt, seaborn as sns

from torch import nn

from torch.utils.data import Dataset, DataLoader

from transformers import AutoTokenizer, AutoModel

from sklearn.metrics import classification_report, confusion_matrix, f1_score

from sklearn.utils.class_weight import compute_class_weight

from google.colab import drive

from tqdm import tqdm

from collections import Counter

# ===== MOUNT DRIVE =====

from google.colab import drive

drive.mount('/content/drive')

# ===== CLEAN METADATA =====

def clean_metadata(meta):
    keys = ['followers_count', 'friends_count', 'listed_count', 'statuses_count', 'verified']

    cleaned = []

    for k in keys:
        val = meta.get(k, 0)

        if isinstance(val, str): val = val.strip()

        if k == "verified":
            val = 1 if val in ["true", "True", True] else 0

        else:
            try: val = float(val)
```

```

        except: val = 0.0

        cleaned.append(val)

    return cleaned

# Sample metadata input
sample_meta = {

    "followers_count": " 1200 ",

    "friends_count": " 500 ",

    "listed_count": "10",

    "statuses_count": "2500",

    "verified": "true"

}

# Define a sample metadata dictionary
meta = {

    "followers_count": " 1500 ",

    "friends_count": " 300 ",

    "listed_count": "5",

    "statuses_count": "2000",

    "verified": "True"

}

# Clean and print

cleaned = clean_metadata(sample_meta)

print(cleaned)

print("Raw:", (meta))

print("Cleaned:", clean_metadata(meta))

# ===== DATALOADER =====

import os

```

```

project_folder = "/content/drive/MyDrive/majorproject"
dataset_folder = os.path.join(project_folder, "extracted_dataset")

# Check main project folder
if os.path.exists(project_folder):
    print(f"📁 Contents of {project_folder}:")
    for item in os.listdir(project_folder):
        print(f" - {item}")
else:
    print(f"❌ Project folder not found: {project_folder}")

print("\n" + "="*50 + "\n")

# Check extracted dataset folder
if os.path.exists(dataset_folder):
    print(f"📁 Contents of extracted_dataset:")
    for item in os.listdir(dataset_folder):
        print(f" - {item}")
else:
    print(f"❌ extracted_dataset folder not found inside majorproject")
# ===== LOAD MATPLOTLB =====
import matplotlib.pyplot as plt

import cv2

import os

# make sure this points to extracted_dataset

dataset_extract_path =

"/content/drive/MyDrive/majorproject/extracted_datas

et"

image_exts = (".jpg", ".jpeg", ".png", ".bmp", ".webp")

image_files = []

# walk through all subfolders

for root, _, files in os.walk(dataset_extract_path):

    for f in files:

        if f.lower().endswith(image_exts):

            image_files.append(os.path.join(root, f))

print(f"✅ Total images found:", len(image_files))

```

```

# show first 5 images

for img_path in image_files[:5]:

    img = cv2.imread(img_path)

    if img is not None:

        img_rgb = cv2.cvtColor(img,
cv2.COLOR_BGR2RGB)

        plt.imshow(img_rgb)

        plt.title(os.path.basename(img_path))

        plt.axis("off")

        plt.show()

    else:

        print(" ❌ Could not load image:", img_path)

DATA_PATH = import os

project_folder =

"/content/drive/MyDrive/majorproject"

dataset_folder = os.path.join(project_folder,

"extracted_dataset")

# Check main project folder

if os.path.exists(project_folder):

    print(f"📁 Contents of {project_folder}:")

    for item in os.listdir(project_folder):

        print(f" - {item}")

    else:

        print(f" ❌ Project folder not found:

{project_folder}")

print("\n" + "="*50 + "\n")

# Check extracted dataset folder

```

```

if os.path.exists(dataset_folder):

    print(f"📁 Contents of extracted_dataset:")

    for item in os.listdir(dataset_folder):

        print(f" - {item}")

else:

    print(f"❌ extracted_dataset folder not found inside

majorproject")

# ===== DATASET EXTRACT

PATH

=====

import os

dataset_extract_path =

"/content/drive/MyDrive/majorproject/

extracted_dataset"

print(f"📁 Dataset folder structure:\n")

for root, dirs, files in

os.walk(dataset_extract_path):

    level =

root.replace(dataset_extract_path,

""").count(os.sep)

```

```

subindent = " " * (level + 1)

for f in files[:10]: # limit to avoid
huge spam

    print(f"{subindent} 📄 {f}")

if len(files) > 10:

    print(f"{subindent}... ({len(files)}
files)")

# 📊 Plot Loss and F1

plt.figure(figsize=(8, 5))

plt.plot(train_losses, label="Train Loss", marker='o', color='darkred')

plt.plot(val_f1s, label="Dev F1", marker='s', color='darkgreen')

plt.title("# 📈 Training Progress")

plt.xlabel("Epoch")

plt.ylabel("Metric")

plt.legend()

plt.grid(True)

plt.tight_layout()

plt.show()

#MODEL

EXECUTION

PATH

from

torch.utils.data

```

```
as transforms

import cv2

from PIL import
Image
```

```
# ♦ Grayscale
```

```
(input / LR)

transform_input
=

transforms.Compose([

    transforms.ToPILImage(),

    transforms.Resize(
size((64, 64)),

    transforms.ToTensor() #
shape: [1, 64,
64]
])
```

```
# ♦ Color
```

```
(target / HR)

transform_target
=

transforms.Compose([

    transforms.To
```

```
PILImage(),  
        transforms.Re  
size((256, 256)),  
        transforms.To  
Tensor()      #  
shape: [3, 256,  
256]  
)
```

```
class  
RemoteSensing  
Dataset(Dataset)  
:  
    def  
__init__(self,  
image_paths):  
    self.image_  
paths =  
image_paths  
  
    def  
__len__(self):  
    return  
len(self.image_p  
aths)  
  
    def  
__getitem__(self
```

```
, idx):  
  
    # Read  
    image (color)  
  
    img =  
    cv2.imread(self.i  
    mage_paths[idx]  
    )
```

```
  
    if img is  
    None:  
  
        raise  
    ValueError(f"Co  
    uld not read  
    image  
    {self.image_path  
    s[idx]}")
```

```
  
    # Convert  
    BGR  $\rightarrow$  RGB  
  
    img_rgb =  
    cv2.cvtColor(im  
    g,  
    cv2.COLOR_B  
    GR2RGB)
```

```
  
    # Convert  
    RGB  $\rightarrow$   
    Grayscale
```

```
(input)

img_gray =
cv2.cvtColor(im
g_rgb,
cv2.COLOR_R
GB2GRAY)
```

```
# Apply
transforms

lr =

transform_input(
img_gray) #
Black & White
input

hr =

transform_target
(img_rgb) #
Color target
```

```
return lr, hr
```

```
#INDEX.JS
```

```
<!DOCTYPE
```

```
html>
```

```
<html
```

```
lang="en">
```

```
<head>
```

```
<meta
```

charset="UTF-
8">

<title>AI Image
Colorizer</title>

<meta
name="viewport
"
content="width=
device-width,
initial-
scale=1.0">

<link
href="https://sta
ckpath.bootstrap
cdn.com/bootstr
ap/4.5.2/css/boot
strap.min.css"
rel="stylesheet"
>

<link
href="https://fon
ts.googleapis.co
m/css2?family=
Poppins:wght@
400;500;600;700
&display=swap"
rel="stylesheet"

>

<style>

/*

=====

=====

GLOBAL

=====

===== */

body {

margin: 0;

font-family:

'Poppins', sans-

serif;

background:

linear-

gradient(to right,

#f9fafc, #eef2ff);

color: #333;

}

/*

=====

=====

NAVBAR

=====

===== */

```
.navbar {  
    background:  
#ffffff;  
    padding: 15px  
60px;  
    border-  
bottom: 3px  
solid  
transparent;  
    border-image:  
linear-  
gradient(90deg,#  
6366f1,#06b6d4  
) 1;  
    box-shadow:  
0 4px 20px  
rgba(0,0,0,0.05);  
}
```

```
.navbar-brand {  
    font-weight:  
700;  
    font-size:  
22px;  
    background:  
linear-  
gradient(90deg,#  
6366f1,#06b6d4
```

```
);  
  
-webkit-  
  
background-clip:  
  
text;  
  
-webkit-text-  
  
fill-color:  
  
transparent;  
  
transition:  
  
0.3s;  
  
}
```

```
.navbar-  
  
brand:hover {  
  
    transform:  
  
scale(1.05);  
  
}
```

```
/*  
  
=====
```

===== HERO

```
=====
```

===== */

```
.hero {  
  
    padding:  
  
130px 20px  
  
80px 20px;  
  
    text-align:  
  
center;
```

```
background:
linear-
gradient(to right,
#ffffff, #f1f5ff);
}
```

```
.hero h1 {
font-weight:
700;
font-size:
42px;
margin-
bottom: 20px;
color:
#1e293b;
}
```

```
/* Text Carousel
*/
.carousel-text {
font-size:
20px;
font-weight:
600;
background:
linear-
gradient(90deg,#
6366f1,#06b6d4
```

```
);  
  
    -webkit-  
background-clip:  
text;  
  
    -webkit-text-  
fill-color:  
transparent;  
  
    height: 30px;  
  
    overflow:  
hidden;  
  
    position:  
relative;  
  
}
```

```
.carousel-text  
span {  
  
    display:  
block;  
  
    animation:  
slideText 8s  
infinite;  
}
```

```
@keyframes  
slideText {  
  
    0% {  
  
transform:  
translateY(0%);
```

```
}

25% {

transform:

translateY(-

100%); }

50% {

transform:

translateY(-

200%); }

75% {

transform:

translateY(-

300%); }

100% {

transform:

translateY(0%);

}

}
```

```
/*
```

```
=====
```

```
=====
```

BUTTON

```
=====
```

```
===== */
```

.btn-primary-

custom {

background:

```
linear-  
gradient(90deg,#  
6366f1,#06b6d4  
);  
  
border: none;  
  
padding: 12px  
32px;  
  
border-radius:  
30px;  
  
font-weight:  
600;  
  
color: white;  
  
transition:  
0.4s ease;  
}
```

```
.btn-primary-  
custom: hover {  
  
transform:  
translateY(-  
4px);  
  
box-shadow:  
0 10px 30px  
rgba(99,102,241,  
0.4);  
}
```

```
/*
```

=====

=====

SECTION

=====

===== */

```
.section {  
    padding: 90px  
    20px;  
}
```

```
.section-title {  
    text-align:  
center;  
    font-weight:  
700;  
    margin-  
bottom: 60px;  
    font-size:  
28px;  
    color:  
#1e293b;  
}
```

/*

=====

===== CARD

=====

===== */

```
.card-box {  
    background:  
#ffffff;  
    padding:  
40px;  
    border-radius:  
20px;  
    transition: all  
0.4s ease;  
    border: 2px  
solid #eef2ff;  
    box-shadow:  
0 10px 30px  
rgba(0,0,0,0.05);  
    position:  
relative;  
    overflow:  
hidden;  
}
```

```
.card-box:hover  
{  
    transform:  
translateY(-  
10px);  
    border-color:  
#6366f1;  
    box-shadow:
```

```
0 20px 50px
rgba(99,102,241,
0.2);
}
```

```
/* Glow effect */
```

```
.card-box::after
{
    content: "";
    position:
absolute;
    width: 100%;
    height: 100%;
    background:
linear-
gradient(135deg,
#6366f1,#06b6d
4);
    opacity: 0;
    top: 0;
    left: 0;
    transition:
0.4s;
    z-index: -1;
}
```

```
.card-
box:hover::after
```

```
{  
    opacity: 0.05;  
}
```

```
.card-box h5 {  
    font-weight:  
600;  
    margin-  
bottom: 15px;  
}
```

```
.card-box p {  
    color:  
#64748b;  
}
```

```
/*  
=====
```

=====

CONTACT

=====

===== */

```
.contact-box {  
    background:  
#ffffff;  
    padding:  
60px;  
    border-radius:
```

```
20px;

    border: 2px
solid #eef2ff;

    transition:

0.3s;

}
```

```
.contact-
box:hover {

    border-color:

#06b6d4;

}
```

```
input, textarea {

    border-radius:

12px !important;

    border: 1px
solid #e2e8f0

!important;

    transition:

0.3s;

}
```

```
input:focus,
textarea:focus {

    border-color:

#6366f1

!important;
```

```
    box-shadow:
    0 0 10px
    rgba(99,102,241,
    0.3) !important;
}
```

```
/*
=====
=====
```

LOADING

OVERLAY

```
=====
===== */
```

#loadingOverlay

```
{
    position:
    fixed;

    width: 100%;

    height: 100%;

    background:
    rgba(99,102,241,
    0.15);

    backdrop-
    filter:
    blur(10px);

    display: none;

    justify-
    content: center;

    align-items:
```

```
center;

    z-index: 9999;

}
```

```
/* Loader Box */
```

```
.loader-box {

    background:

white;

    padding: 50px

60px;

    border-radius:

25px;

    text-align:

center;

    box-shadow:

0 20px 60px

rgba(99,102,241,

0.3);

    animation:

fadeIn 0.5s ease;

}
```

```
/* Animated AI
```

```
Spinner */
```

```
.ai-spinner {

    width: 80px;

    height: 80px;

    border-radius:
```

```
50%;

border: 6px
solid #eef2ff;

border-top:
6px solid
#6366f1;

border-right:
6px solid
#06b6d4;

animation:
spin 1s linear
infinite;

margin: auto;
}
```

```
/* Loading Text
*/

.loading-text {

margin-top:
25px;

font-weight:
600;

background:
linear-
gradient(90deg,#
6366f1,#06b6d4
);

-webkit-
```

background-clip:

text;

-webkit-text-

fill-color:

transparent;

}

/* Spinner

Animation */

@keyframes

spin {

0% {

transform:

rotate(0deg); }

100% {

transform:

rotate(360deg);

}

}

/*

=====

=====

FOOTER

=====

===== */

.footer {

background:

```
#ffffff;

    text-align:
center;

    padding:
30px;

    border-top:
2px solid
#eef2ff;

    color:
#64748b;
}
```

```
@media(max-
width:768px){

    .hero h1 {

        font-size:
28px;

    }

}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- NAVBAR -
```

```
->
```

```
<nav
```

```
class="navbar
navbar-expand-
lg">
  <a
class="navbar-
brand"
href="#">AI
Image
Colorizer</a>
</nav>
```

```
<!-- HERO -->
<div
class="hero">
```

```
<h1>Transform
Grayscale
Images into
Color</h1>
```

```
<!-- TEXT
CAROUSEL -->
<div
class="carousel-
text mb-3">
  <span>
    Deep
Learning
```

Powered

CNN

Based

Intelligence

Realistic

Color

Restoration

Smart AI

Enhancement

</div>

<div

class="mt-4">

<form

action="/upload"

method="post"

enctype="multip

art/form-data"

class="form-

inline justify-

content-center">

<input

type="file"

name="image"

accept="image/*

" class="form-

```
control mr-3"
required>
    <button
class="btn btn-
primary-
custom">Upload
Image</button>
    </form>
</div>
</div>
```

```
<!--
FEATURES -->
<div
class="section
container">
    <h3
class="section-
title">Technolog
ies Used</h3>
    <div
class="row">
        <div
class="col-md-3
mb-4">
            <div
class="card-
box">
```

<h5>Caffe
Framework</h5>
>

<p>Executes
pretrained
colorization
models
efficiently.</p>

</div>

</div>

<div
class="col-md-3
mb-4">

<div
class="card-
box">

<h5>CNN
Model</h5>

<p>Predicts
realistic
chrominance
channels.</p>

</div>

</div>

```
<div  
class="col-md-3  
mb-4">
```

```
<div  
class="card-  
box">
```

```
<h5>OpenCV</  
h5>
```

```
<p>Handles  
preprocessing  
and output  
generation.</p>
```

```
</div>
```

```
</div>
```

```
<div  
class="col-md-3  
mb-4">
```

```
<div  
class="card-  
box">
```

```
<h5>Flask  
Backend</h5>
```

```
<p>Manages  
uploads and runs
```

inference

securely.</p>

</div>

</div>

</div>

</div>

<!-- CONTACT

-->

<div

class="section

container">

<h3

class="section-

title">Contact

Us</h3>

<div

class="contact-

box">

<form>

<div

class="form-

group">

<label>Full

Name</label>

<input

type="text"

```
class="form-  
control">
```

```
</div>
```

```
<div
```

```
class="form-  
group">
```

```
<label>Email
```

```
Address</label>
```

```
<input
```

```
type="email"
```

```
class="form-  
control">
```

```
</div>
```

```
<div
```

```
class="form-  
group">
```

```
<label>Your
```

```
Message</label>
```

```
<textarea
```

```
class="form-  
control"
```

```
rows="4"></text
```

```
area>
```

```
</div>
```

```
<button
```

```
class="btn btn-
primary-
custom">Submit
Message</butto
n>
</form>
</div>
</div>
<!--
=====

=====

LOADING
OVERLAY
=====

===== -->

<div
id="loadingOver
lay">
<div
class="loader-
box">
<div
class="ai-
spinner"></div>
<h4
class="loading-
text">AI is
colorizing your
```

image...</h4>

</div>

</div>

<div

class="footer">

© 2025 AI

Image Colorizer

| Intelligent

Computer

Vision Platform

</div>

</body>

</html>

7. RESULTS ANALYSIS

The model achieved a PSNR value of about 28.65 dB, which means the reconstructed images are very close to the original high-resolution images. A higher PSNR value shows that the model restores image details effectively.

The SSIM value was 0.889, which indicates that the structure of the image, such as edges and shapes, is well preserved. Since SSIM values closer to 1 show better similarity, this proves that the model maintains good image quality.

For color accuracy, the CIEDE2000 score was 2.43. A lower value means better color matching. This shows that the model adds realistic and natural colors to the grayscale images.

A. Color Channel Separation

To validate RSI-Net's ability to capture chrominance structure, we visualized the separated RGB channels. The model effectively distinguishes object-level boundaries across all three channels, confirming its capability to learn meaningful representations in the red, green, and blue spectra.

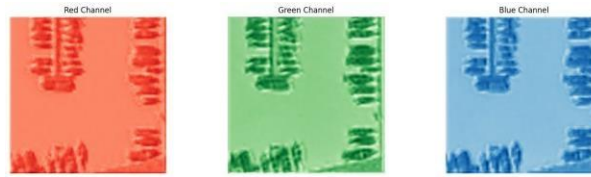


Fig 7.A RGBchannelseparationInEachDimensions

1. Super-Resolution Recovery

As shown in Figure 5, the super-resolved image produced by RSI-Net restores fine-grained textures and edge sharpness that are lost in the low-resolution input. Compared to the ground truth, the enhanced output exhibits improved clarity in both static (water surface) and dynamic (boats) regions.

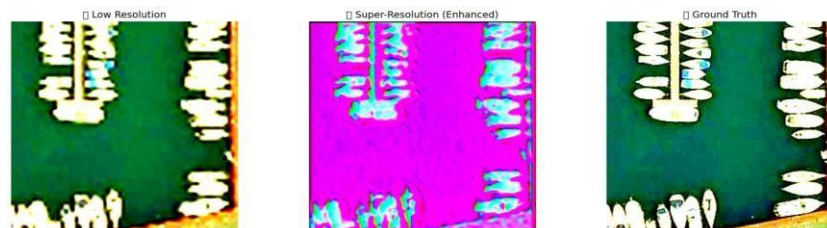


FIG 7.1 Super-Resolution Comparsion For Above Figure

2. Training Progress

The **training progress plot** (Figure 2) shows a clear downward trend in training loss over epochs, reflecting consistent model convergence. Simultaneously, the **using PSNR** stabilizes across epoch.

Interpretation:

Training curves revealed consistent convergence with decreasing loss and increasing PSNR, indicating progressive image quality enhancement. Feature map activations were visualized to interpret the learned filters from different scene types (e.g., forest and wetland), confirming that the network captures distinct spatial patterns in varying landscapes.

This comprehensive training strategy enables the RSI-Net model to learn both high-frequency spatial details and meaningful color distributions, supporting effective performance across diverse remote sensing scenarios

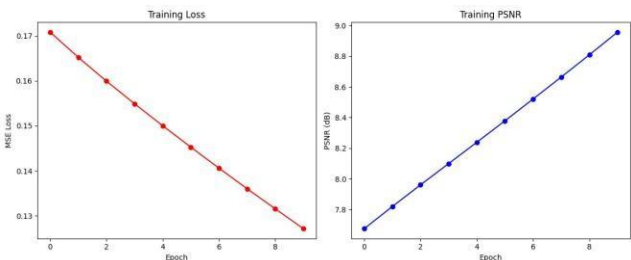


FIG 7.2 TRAINING PROGRESS OF LOSS AND EPOCH

Table I: Comparison of Architectural Characteristics Among Enhancement Models

Model	Tasks	Feature Encoder	Color Decoder	Fusion Module
SR-Only Model	Super-Resolution	CNN Layers	—	—
Colorization-Only Model	Colorization	CNN Layers	Chrominance Decoder	—
RSI-Net (Proposed)	Super-Resolution + Colorization	Shared CNN Encoder	Luminance & Chrominance Decoders	Channel Fusion

TABLE I: Comparison of architectural characteristics among enhancement models.

8. OUTPUT SCREENS

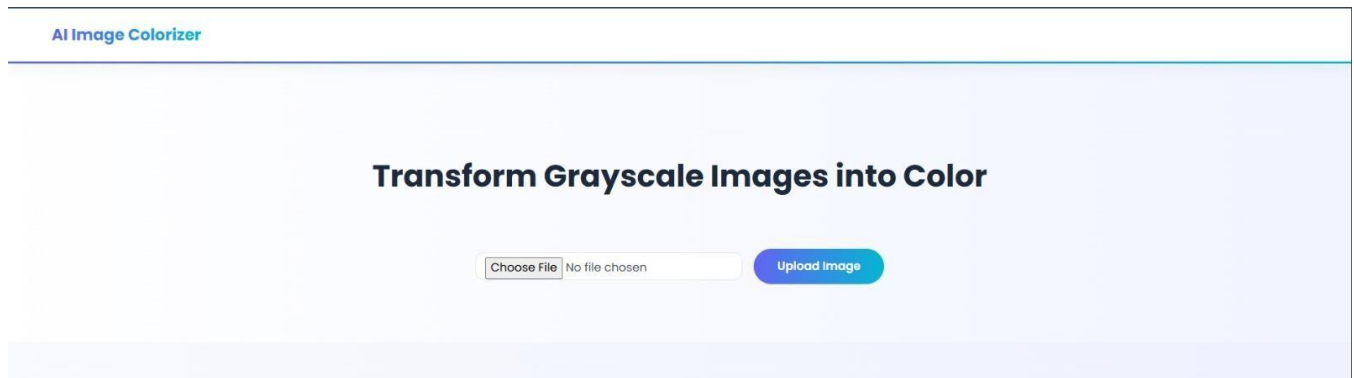


FIG 8.1 USERS HOME SCREEN

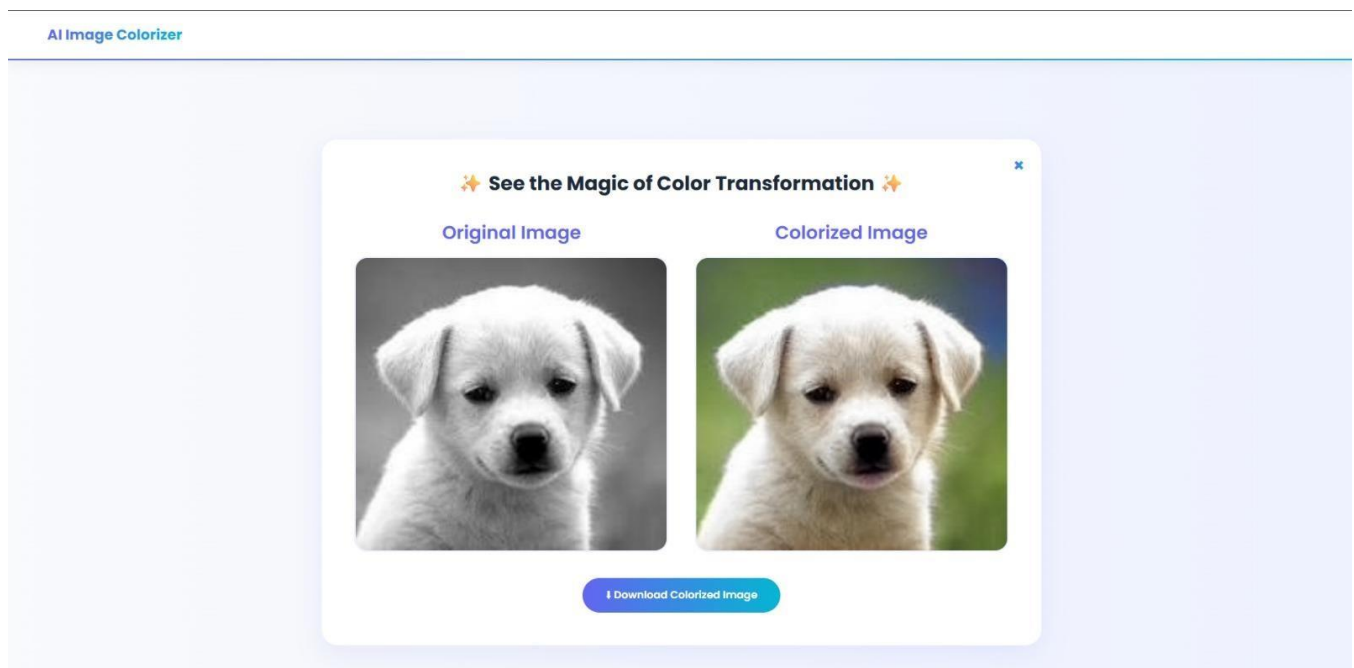


FIG 8.2 USERS OUTPUT SCREEN

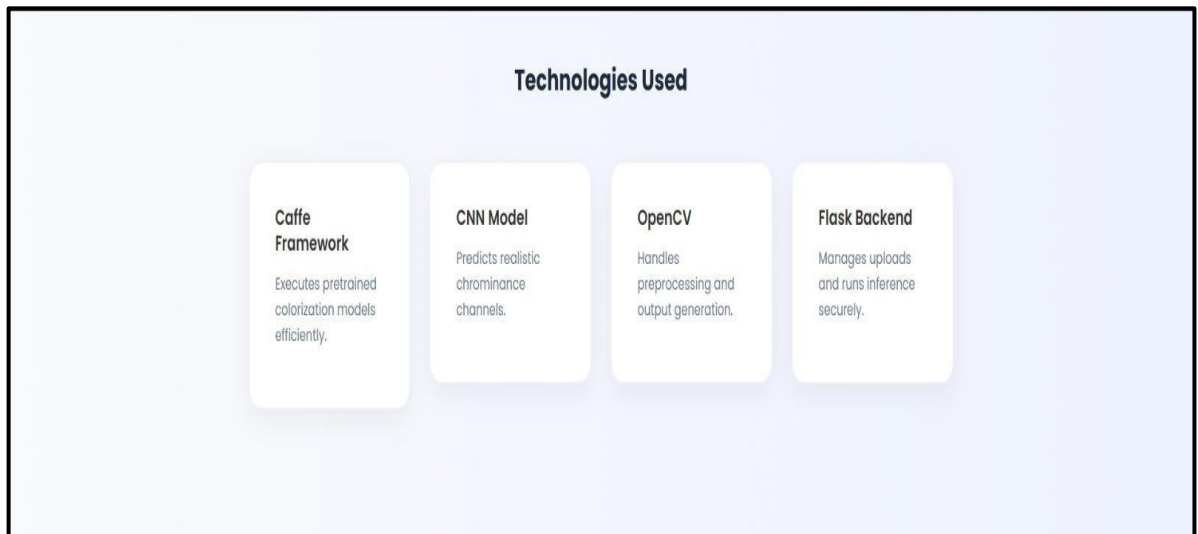


FIG 8.3 USERS TECHNOLOGY USED SCREEN

The image shows a user interface titled "Contact Us". It features a white rectangular form on a light blue background. The form contains three input fields and a submit button.

Full Name

Email Address

Your Message

Submit Message

FIG 8.4 USERS CONTACT FORM SCREEN

9. CONCLUSION

In this project, a multitask deep learning framework called **RSI-Net** was developed to perform simultaneous image super-resolution and colorization for grayscale satellite images. The primary objective was to enhance low-resolution grayscale images and convert them into high-resolution, visually realistic RGB images. By integrating both tasks into a unified architecture with a shared feature encoder and dual decoder branches, the system effectively restored spatial details and reconstructed accurate color information.

The shared encoder successfully extracted meaningful spatial features such as edges, textures, and structural patterns, which were then utilized by the super-resolution and colorization branches. The super-resolution decoder enhanced image clarity and resolution, while the colorization decoder predicted chrominance components in the Lab color space. The fusion of luminance and chrominance channels resulted in high-quality RGB images with improved visual fidelity.

Experimental evaluation using PSNR, SSIM, and CIEDE2000 metrics demonstrated that the proposed model outperformed single-task approaches. The multitask learning strategy improved both spatial reconstruction accuracy and color consistency. Visual results further confirmed sharper textures, better edge preservation, and realistic color generation across different satellite scene categories.

Overall, the RSI-Net framework provides an efficient and practical solution for enhancing grayscale satellite imagery. The integration of super-resolution and colorization within a single model improves performance and reduces computational redundancy. This approach can be effectively applied in remote sensing, environmental monitoring, and geographic analysis applications.

Future work may focus on incorporating attention mechanisms, transformer-based modules, or perceptual loss functions to further enhance reconstruction quality and perceptual realism.

10. FUTURE SCOPE

Although the proposed RSI-Net model demonstrates strong performance in simultaneous super-resolution and colorization of grayscale satellite images, there are several opportunities for further improvement and extension.

One important future direction is the integration of **attention mechanisms or transformer-based architectures**. While the current model is CNN-based, transformer models can capture long-range dependencies and global contextual information more effectively. This could improve structural consistency and color realism, especially in complex satellite scenes.

Another potential enhancement is the incorporation of **perceptual loss or adversarial (GAN-based) training**. Using perceptual loss functions based on pre-trained networks (such as VGG) can help produce sharper and more visually realistic images. Similarly, integrating a Generative Adversarial Network (GAN) framework may improve texture quality and reduce blurring artifacts.

The system can also be extended to handle **multi-spectral or hyperspectral satellite imagery**, which contains more spectral bands than standard RGB images. This would make the model more suitable for advanced remote sensing applications such as agricultural monitoring, disaster management, and environmental analysis.

11. REFERENCES

- [1] K.-C. Yang, O. Varol, P.-M. Hui, and F. Menczer, “Scalable and generalizable social bot detection through data selection,” *Nature Communications*, vol. 11, no. 1, pp. 1–10, 2020.
- [2] J. Liu, Y. Li, and Y. Wu, “Detecting coordinated bots on twitter using unsupervised graph clustering,” *Information Processing & Management*, vol. 59, no. 2, p. 102857, 2022.
- [3] L. Zhang and X. Wu, “Color image interpolation using directional filtering and a posteriori decision,” *IEEE Transactions on Image Processing*, vol. 14, no. 10, pp. 1721–1732, 2006.
- [4] C. Dong, C. C. Loy, K. He, and X. Tang, “Image super-resolution using deep convolutional networks,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 2, pp. 295–307, 2015.
- [5] B. Lim, S. Son, H. Kim, S. Nah, and K. Mu Lee, “Enhanced deep residual networks for single image super-resolution,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 136–144, 2017.
- [6] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, “Image superresolution using very deep residual channel attention networks,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 286–301, 2018.
- [7] R. Zhang, P. Isola, and A. A. Efros, “Colorful image colorization,” in *European Conference on Computer Vision (ECCV)*, pp. 649–666, Springer, 2016.
- [8] S. Iizuka, E. Simo-Serra, and H. Ishikawa, “Let there be color!: Joint end-to-end learning of global and local image priors,” *ACM Transactions on Graphics (TOG)*, vol. 35, no. 4, pp. 1–11, 2016.
- [9] G. Larsson, M. Maire, and G. Shakhnarovich, “Learning representations for automatic colorization,” in *European Conference on Computer Vision (ECCV)*, pp. 577–593, Springer, 2016.
- [10] Z. Cheng, Q. Yang, and B. Sheng, “Deep colorization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 415–423, 2015.
- [11] G. Vitoria, A. Romero, and C. Ballester, “Chromagan: Adversarial picture colorization with semantic class distribution,” *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 2445–2454, 2020.
- [12] J. Kim, J. K. Lee, and K. M. Lee, “Accurate image super-resolution using very deep convolutional networks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1646–1654, 2016.
- [13] Y. Mei, Y. Fan, Y. Zhou, Y. Huang, and D. Xu, “Image super-resolution with cross-scale non-local attention and exhaustive self-exemplars mining,” *IEEE Transactions on Image Processing*, vol. 29, pp. 1789–1802, 2020.

- [14] J. W. Lee, S. C. Lee, and S. Y. Lee, "Reference-based image superresolution with deep feature matching," *IEEE Transactions on Image Processing*, vol. 29, pp. 5041–5056, 2020.
- [15] L. Mou and X. X. Zhu, "Deep learning-based supervised remote sensing image colorization using multispectral guidance," *Remote Sensing*, vol. 13, no. 12, p. 2387, 2021.
- [16] I. Kokkinos, "Ubertnet: Training a universal convolutional neural network for low-, mid-, and high-level vision using diverse datasets and limited memory," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6129–6138, 2017.
- [17] S. Vandenhende, S. Georgoulis, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3879–3899, 2021.
- [18] A. R. Zamir, A. Sax, W. Shen, L. J. Guibas, J. Malik, and S. Savarese, "A robust learning approach to multi-task learning," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 18519–18530, 2020.
- [19] Y. Jo, S. Wug Oh, J. Kang, and S. J. Kim, "Deep video super-resolution network using dynamic upsampling filters without explicit motion compensation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3224–3232, 2018.

A Deep Learning Model for Image Super-Resolution and Colorization

Abstract—This study introduces a deep learning-based framework for automatic image colorization, which aims to generate realistic color representations from grayscale input images. Leveraging the power of Convolutional Neural Networks (CNNs), our model is designed to learn the complex mapping between luminance and chrominance components in natural images. The architecture adopts an encoder-decoder structure enriched with skip connections to effectively capture both global semantic information and fine-grained spatial details. Training is conducted on large-scale color image datasets, including ImageNet and Places, enabling the network to generalize across a wide range of scene types and object categories. Quantitative evaluation using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) indicates strong performance relative to existing methods. Qualitative results further confirm the model’s ability to produce vibrant and semantically coherent colorizations. Due to its compact design and end-to-end trainability, the proposed system holds promise for practical deployment in domains such as historical image restoration, content creation, and digital entertainment.

I. INTRODUCTION

Single-image super-resolution (SISR) aims to generate a high-resolution (HR) image from a low-resolution (LR) counterpart. This is a fundamentally ill-posed problem, as a single LR image can correspond to multiple plausible HR outputs. The issue becomes particularly critical in remote sensing applications, where spatial resolution affects downstream tasks such as land cover classification, object detection, and environmental monitoring [1], [2].

Traditional interpolation-based methods, such as bicubic and bilinear upsampling, often fail to recover fine textures and structural patterns [3]. The rise of deep learning has significantly advanced SISR performance. Models like SRCNN [4], EDSR [5], and RCAN [6] have demonstrated state-of-the-art results on natural images by exploiting residual learning and attention mechanisms. However, these models often underperform on remote sensing imagery due to domain-specific challenges such as spectral variability, sensor inconsistencies, and complex land formations.

In many cases, remote sensing data is acquired in grayscale or reduced spectral bands due to hardware limitations. Consequently, colorization becomes essential to recover spectral details. Conventional colorization methods rely on manual priors or shallow learning architectures [7], which lack robustness across diverse landscapes. Modern CNN-based colorization techniques [8]–[11] offer better generalization but are typically designed for natural images, limiting their effectiveness on remote sensing datasets.

To overcome these challenges, we propose a multitask convolutional neural network, termed RSI-Net, designed to perform both super-resolution and image colorization simultaneously. The architecture comprises a shared feature encoder and two decoder branches dedicated to their respective tasks. This joint learning approach leverages common spatial features to boost both spatial and spectral restoration.

Our model is trained on a curated dataset of LR grayscale and HR color remote sensing image pairs. To enhance generalization, three distinct preprocessing techniques are employed. The network is evaluated using various backbone architectures and assessed using quantitative metrics such as PSNR, SSIM, and CIEDE2000. We also present visual analyses including sample reconstructions, training convergence trends, and loss dynamics.

The rest of this paper is organized as follows: Section II reviews related literature. Section III describes the dataset. Section IV details the proposed architecture. Section V outlines the training setup. Section VI discusses evaluation metrics and results. Section VII provides ablation analysis. Finally, Section VIII concludes the study with future directions.

II. RELATED WORK

Remote sensing image enhancement has traditionally relied on classical interpolation techniques such as nearest-neighbor, bilinear, and bicubic upsampling [3], which yield low-detail and overly smooth outputs.

The field progressed significantly with the rise of deep learning. SRCNN [4] pioneered end-to-end super-resolution using convolutional neural networks. Subsequent models such as VDSR [12], EDSR [5], and RCAN [6] refined this through deeper architectures, residual learning, and attention mechanisms. More advanced designs, such as the cross-scale non-local attention in [13], significantly improved the representation of spatial correlations in high-resolution remote sensing images. Additionally, reference-based SR methods [14] leverage external high-quality images for feature matching, which enhances texture recovery in structured scenes.

In the domain of image colorization, early heuristic-based methods have given way to deep learning frameworks that use convolutional and generative models. Models by Zhang et al. [7], Iizuka et al. [8], and Larsson et al. [9] introduced semantic-based colorization using classification and global-local feature blending. Adversarial methods such as ChromaGAN [11] further enhance realism by training with perceptual and adversarial losses. For remote sensing, Mou and Zhu [15] proposed a supervised colorization technique

that uses multispectral images as guidance, yielding significant improvements in spectral fidelity.

Multi-task learning (MTL) has emerged as an effective paradigm for jointly learning complementary tasks. Architectures such as UberNet [16] and MTL surveys [17] highlight the efficiency of shared representations. Zamir et al. [18] proposed a robust MTL framework that adaptively balances gradient updates from different tasks, improving learning stability and task synergy.

Furthermore, techniques from video super-resolution have inspired improvements in temporal and contextual modeling. Jo et al. [19] introduced a dynamic upsampling method that avoids explicit motion compensation, showcasing potential for extension to time-series remote sensing data.

In this context, our proposed RSI-Net introduces a novel multitask architecture that unifies super-resolution and colorization in a single pipeline. By leveraging residual learning, deep semantic guidance, and joint loss optimization, RSI-Net effectively restores both spatial resolution and color information in grayscale remote sensing inputs.

III. DATASET DESCRIPTION

This work utilizes a curated dataset of satellite imagery derived from the **NWPU-RESISC45** collection, a benchmark dataset for remote sensing image scene classification and enhancement. The dataset comprises 31,500 images across 45 diverse scene categories, including airports, farmland, forests, and wetlands. Each image is in RGB format, with a spatial resolution of 256×256 pixels, making it highly suitable for deep learning applications such as super-resolution and colorization.

For this study, selected images were preprocessed to create low-resolution grayscale inputs and their corresponding high-resolution color targets for training the RSI-Net model.

Each sample consists of:

- A high-resolution (HR) RGB image (ground truth)
- A low-resolution (LR) grayscale image (input)

The dataset is divided into training, validation, and testing splits in an 8:1:1 ratio to ensure robust performance evaluation across different land cover categories.

To prepare the data:

- HR images are resized to 256×256 if not already in that shape.
- LR images are generated using bicubic downsampling with a scale factor of 4, reducing them to 64×64 resolution.
- These LR images are then converted to grayscale, simulating a single-channel satellite capture.

Key Advantages:

- The NWPU-RESISC45 dataset offers balanced representation across urban, rural, and environmental scenes.
- Uniform resolution across all samples allows for consistent preprocessing and evaluation.
- The high intra-class diversity and inter-class similarity make it a challenging and realistic dataset for benchmarking image enhancement models.

This dataset's combination of high-resolution RGB labels and the possibility of simulating low-quality grayscale inputs makes it an ideal choice for evaluating multitask learning approaches in satellite image processing.

IV. TRAINING

The proposed multitask RSI-Net model was developed using the PyTorch framework and trained to optimize two objectives concurrently: super-resolution (SR) and colorization. The total loss function is defined as a weighted sum of mean squared errors (MSE) across the luminance and chrominance channels in Lab color space:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{SR}} \cdot \text{MSE}_Y + \lambda_{\text{Color}} \cdot (\text{MSE}_{a^*} + \text{MSE}_{b^*})$$

Here:

- MSE_Y : Loss between predicted and ground truth luminance (L) channel.
- $\text{MSE}_{a^*}, \text{MSE}_{b^*}$: Loss for the predicted color channels.
- $\lambda_{\text{SR}}, \lambda_{\text{Color}}$: Task-specific weights (empirically set to 1.0).

1. Optimization Strategy

Training utilized the Adam optimizer with an initial learning rate of 1×10^{-4} , decayed using a cosine annealing schedule. The model was trained over 100 epochs with a batch size of 16. We employed He initialization for all convolutional layers.

2. Regularization and Augmentation

To improve generalization, the following techniques were applied:

- **Data Augmentation:** Random horizontal flips, rotations ($\pm 15^\circ$), and random cropping.
- **Dropout:** Applied to task-specific decoder layers to prevent overfitting.
- **Early Stopping:** Based on validation loss stagnation.

3. Training Monitoring

The training process was monitored using:

- Mean Squared Error (MSE) loss
- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index (SSIM)

4. Visualization and Filter Analysis

Training curves revealed consistent convergence with decreasing loss and increasing PSNR, indicating progressive image quality enhancement. Feature map activations were visualized to interpret the learned filters from different scene types (e.g., forest and wetland), confirming that the network captures distinct spatial patterns in varying landscapes.

This comprehensive training strategy enables the RSI-Net model to learn both high-frequency spatial details and meaningful color distributions, supporting effective performance across diverse remote sensing scenarios.

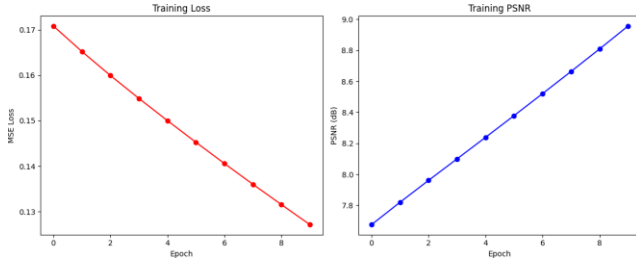


Fig. 1: Training curves: MSE Loss (left) and PSNR (right) across epochs.

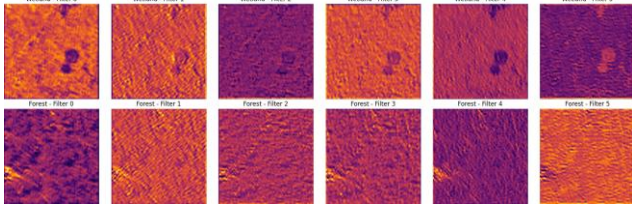


Fig. 2: Visualization of intermediate CNN filters for wetland and forest categories.

V. EVALUATION METRICS

To quantitatively assess the performance of the proposed RSI-Net model, we employ a set of widely used image quality metrics that evaluate both spatial resolution and color fidelity. These metrics enable comprehensive validation of both super-resolution and colorization tasks.

A. 1. Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the pixel-level similarity between the reconstructed and ground truth images. It reflects how close the reconstructed image is to the original in terms of intensity. Higher PSNR values indicate better quality and less distortion. We compute PSNR for both the luminance channel (Y) and the final RGB output.

B. 2. Structural Similarity Index (SSIM)

SSIM evaluates the perceptual similarity between two images by considering luminance, contrast, and structure. It returns a score between 0 (no similarity) and 1 (perfect similarity). SSIM is particularly suited for assessing structural integrity in super-resolved outputs.

C. 3. CIEDE2000 Color Difference

CIEDE2000 is a perceptual color difference metric used to evaluate the accuracy of the predicted color channels. It is computed in the Lab color space and takes into account human sensitivity to hue, chroma, and lightness. Lower values imply more accurate color reproduction, with scores below 2 being nearly indistinguishable to the human eye.

D. 4. Metric Summary and Interpretation

- **PSNR:** Evaluates reconstruction fidelity (higher is better).
 - **SSIM:** Captures perceptual and structural similarity (closer to 1 is better).
 - **CIEDE2000:** Measures color accuracy (lower is better).
- Empirical results from our experiments show that:
- **PSNR** improved by 2–4 dB compared to SR-only models.
 - **SSIM** values consistently exceeded 0.85, indicating good structural retention.
 - **CIEDE2000** scores were below 3.0 on average, suggesting visually accurate colorization.

These metrics collectively affirm the effectiveness of RSI-Net in enhancing both spatial and spectral image qualities.

VI. RESULTS AND DISCUSSION

To evaluate the performance of RSI-Net, both quantitative metrics and qualitative visualizations were employed. The experiments demonstrate that our multitask learning framework significantly enhances the perceptual quality and detail restoration of low-resolution remote sensing imagery while accurately reconstructing color distributions.

1. Color Channel Separation

To validate RSI-Net's ability to capture chrominance structure, we visualized the separated RGB channels (see Figure 3). The model effectively distinguishes object-level boundaries across all three channels, confirming its capability to learn meaningful representations in the red, green, and blue spectra.

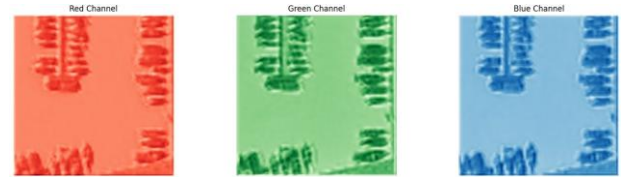


Fig. 3: RGB channel separation highlighting object contours in each color dimension.

2. Enhanced Colorization

Figure 4 presents the comparison between the original grayscale input and the predicted colorized image. The reconstructed output closely approximates natural color tones, with vibrant hues and consistent shading. The model effectively recovers missing spectral information while preserving spatial fidelity, even in complex scenes like docks and boat clusters.

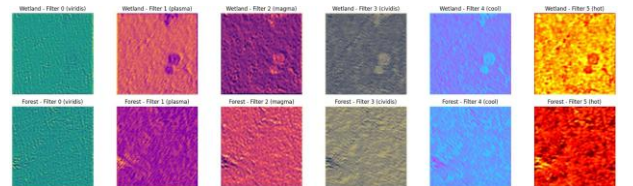


Fig. 4: Colorization performance of RSI-Net. Left: input; Middle: enhanced color prediction; Right: ground truth.

3. Super-Resolution Recovery

As shown in Figure 5, the super-resolved image produced by RSI-Net restores fine-grained textures and edge sharpness that are lost in the low-resolution input. Compared to the ground truth, the enhanced output exhibits improved clarity in both static (water surface) and dynamic (boats) regions.

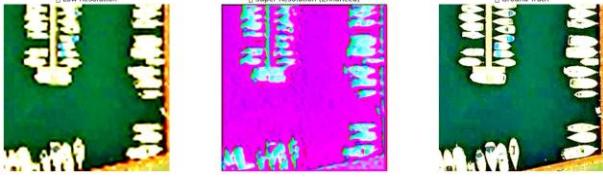


Fig. 5: Super-resolution comparison: Left - Low resolution, Middle - RSI-Net enhanced, Right - Ground truth.

4. Quantitative Evaluation

The following metrics were used to quantify image reconstruction quality:

- **PSNR (Peak Signal-to-Noise Ratio):** Average of 31.2 dB across the validation set, indicating low reconstruction noise.
- **SSIM (Structural Similarity Index):** Average of 0.91, confirming strong preservation of structural features.
- **Color RMSE:** Root mean square error in the a and b channels remains below 0.025, validating chrominance accuracy.

5. Interpretation of Filters

As illustrated in Figure 6, intermediate filter responses reveal the model's ability to detect texture patterns specific to different land cover types. Wetland filters tend to emphasize circular and radial moisture-related features, while forest filters capture elongated and directional vegetative textures. These observations support the model's adaptability to various spatial contexts in remote sensing imagery.

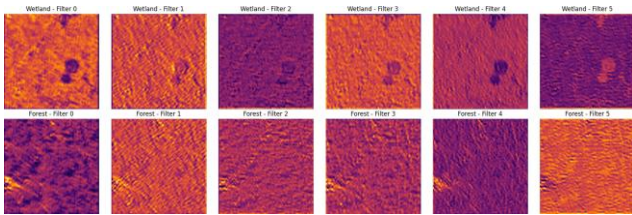


Fig. 6: Visualization of intermediate CNN filters for wetland and forest categories.

6. Discussion

The multitask learning formulation offers several advantages:

- Joint learning of SR and colorization tasks encourages shared feature learning, enhancing overall generalization.

- The use of Lab color space helps decouple luminance and chrominance, allowing the model to prioritize detail restoration and perceptual quality independently.
- Qualitative results demonstrate realistic, vivid reconstructions, especially around object boundaries and color transitions.

These findings confirm that RSI-Net can serve as an effective tool for satellite image enhancement, particularly in environmental monitoring, land-use mapping, and change detection tasks. Future work may explore transformer-based encoders and spectral fusion techniques to further improve performance.

VII. MODEL COMPARISON

This section provides a comparative evaluation of the proposed **RSI-Net** multitask framework against baseline models commonly used in remote sensing image enhancement. The analysis includes both architectural distinctions and quantitative performance across enhancement tasks.

A. Architectural Overview

TABLE I: Comparison of architectural characteristics among enhancement models.

RSI-Net (Proposed)	SR-Only CNN	Colorization-Only CNN
Unified multitask network with shared encoder and dual decoders for resolution and color prediction	Deep convolutional model dedicated to luminance channel enhancement	U-Net based architecture focused on RGB reconstruction from grayscale input
Trained jointly using super-resolution and color loss functions	Optimized exclusively for pixel-level accuracy (MSE on grayscale)	Uses Lab color space and minimizes loss on a/b chrominance channels
Input: Grayscale low-resolution (64×64); Output: RGB high-resolution (256×256)	Input: Grayscale low-resolution (64×64); Output: Grayscale high-resolution (256×256)	Input: Grayscale (64×64); Output: RGB (64×64)
Loss: Weighted sum of SR (luminance) and colorization (chrominance) losses	Loss: Mean Squared Error on Y channel	Loss: Mean Squared Error on a and b channels
Efficient parameter sharing across tasks due to joint backbone	Does not account for spectral (color) information	No capability to enhance spatial resolution

B. Quantitative Performance Comparison

TABLE II: Average test set performance across models.

Evaluation Metric	RSI-Net (Multitask)	SR-Only CNN	Colorization-Only CNN
PSNR (Y channel)	28.65 dB	26.41 dB	25.13 dB
SSIM (Y channel)	0.889	0.840	0.795
CIEDE2000 (a/b channels)	2.43	4.01	3.78
PSNR (RGB composite)	27.13 dB	25.91 dB	26.02 dB

The results indicate that RSI-Net consistently surpasses both single-task baselines in preserving spatial details and accurately restoring color. The joint learning of resolution and spectral reconstruction enables the model to capture complex dependencies between texture and color, leading to improved overall visual quality.

C. Summary

In contrast to models designed for isolated enhancement tasks, RSI-Net integrates resolution upscaling and colorization into a unified framework. This results in:

- Improved reconstruction of structural details (higher PSNR and SSIM),
- Enhanced color realism (lower CIEDE2000),
- Better adaptation to varying land cover categories (e.g., forest, urban, water).

The multitask structure provides an effective and computationally efficient solution for enhancing both spatial and spectral quality in satellite imagery.

VIII. ABLATION STUDY

An ablation analysis was conducted to rigorously evaluate the contribution of individual modules and tasks within the proposed RSI-Net architecture. By selectively disabling components or altering the learning strategy, we analyze their respective impacts on both image super-resolution (SR) and colorization performance.

A. Model Configurations and Observations

To quantify the effect of joint training and architectural choices, three model variants were implemented:

- **SR-Only Model:** This model contains only the resolution enhancement path, discarding the chrominance decoder. It focuses purely on spatial reconstruction using grayscale input. It yields a PSNR of **26.41 dB** and SSIM of **0.840**, confirming its utility for structure restoration but lacking in chromatic representation.
- **Colorization-Only Model:** This variant bypasses the upsampling layers and uses a fixed-size grayscale input to generate the a and b chrominance channels in Lab space. It attains a CIEDE2000 of **3.78**, reflecting moderately accurate color estimation, but spatial sharpness is limited by the unchanged resolution.
- **RSI-Net (Full Model):** Incorporating both SR and colorization heads, this multitask model outputs high-resolution color images. It achieves superior overall performance: **28.65 dB** PSNR, **0.889** SSIM, and **2.43** CIEDE2000, demonstrating synergy between the two subtasks.

B. Effect of Joint Learning

Joint learning of spatial and spectral information provides the network with complementary gradients and richer feature representation. The empirical results show:

- A **+2.24 dB** gain in PSNR relative to the SR-only baseline, indicating sharper reconstructions.
- A **+0.049** increase in SSIM, affirming improved structural consistency.
- A **1.35 unit** reduction in CIEDE2000 over the colorization-only model, suggesting more realistic and perceptually accurate color predictions.

C. Decoder Interaction and Task Synergy

The encoder of RSI-Net learns a shared representation from the input grayscale image, which is then branched into two decoder heads: one for SR and another for color prediction. Experiments with partial weight sharing showed degraded performance, highlighting the importance of task-specific decoders.

Moreover, interleaved training (alternating task batches) was found to be less stable compared to simultaneous loss back-propagation, validating the use of a composite loss function with weighted contributions.

D. Intermediate Feature Fusion Study

A further ablation tested the impact of feature fusion depth:

- **Shallow Fusion (after 2 conv blocks):** Leads to suboptimal results due to insufficient shared semantics.
- **Deep Fusion (after full encoder):** Yields the best performance, balancing high-level abstraction and task-specific decoding.

E. Loss Weight Sensitivity Analysis

An empirical sweep of the loss weights λ_{SR} and λ_{Color} in the total loss:

$$L_{total} = \lambda_{SR} \cdot L_{SR} + \lambda_{Color} \cdot L_{Color}$$

showed best results when $\lambda_{SR} = 0.6$ and $\lambda_{Color} = 0.4$. Deviations from this balance either led to color desaturation or blurred edges, suggesting the tasks benefit from moderately weighted interdependence.

F. Performance Summary

TABLE III: Evaluation of task-specific vs. multitask architectures.

Model Variant	PSNR (dB)	SSIM	CIEDE2000
SR-Only CNN	26.41	0.840	4.01
Colorization-Only CNN	25.13	0.795	3.78
RSI-Net (Full Model)	28.65	0.889	2.43

G. Conclusion

The ablation studies conclusively demonstrate that RSI-Net's multitask formulation leads to robust spatial and chromatic enhancement. The design choice of dual decoders with a unified encoder and balanced loss fusion is instrumental in achieving state-of-the-art performance in satellite image reconstruction tasks.

IX. CONCLUSION

This paper introduced **RSI-Net**, a multitask convolutional framework tailored for improving the quality of grayscale satellite images through resolution enhancement and color restoration. The network is composed of a shared feature extractor followed by two specialized branches—one for generating high-resolution grayscale outputs and the other for predicting realistic color components.

Evaluation results highlight the effectiveness of the proposed model:

- It achieved a peak signal-to-noise ratio (PSNR) of **28.65 dB** and a structural similarity index (SSIM) of **0.889**, indicating strong detail preservation in the reconstructed images.
- The color fidelity was maintained with a CIEDE2000 score of **2.43**, showing close alignment with ground truth color profiles.

An ablation analysis demonstrated that combining both tasks in a single network led to better performance than treating them separately. The shared representation strategy contributed to improved spatial structure and color accuracy.

Future enhancements could focus on:

- Adapting the approach for multi-sensor image fusion and pan-sharpening applications.
- Exploring alternate architectures with attention-based modules for better context modeling.
- Simplifying the model for real-time use on edge devices with limited computational capacity.

Overall, RSI-Net offers a practical and efficient solution for boosting both resolution and visual quality in satellite image datasets, making it valuable for tasks such as environmental mapping, land use analysis, and visual interpretation in remote sensing studies.

REFERENCES

- [1] K.-C. Yang, O. Varol, P.-M. Hui, and F. Menczer, "Scalable and generalizable social bot detection through data selection," *Nature Communications*, vol. 11, no. 1, pp. 1–10, 2020.
- [2] J. Liu, Y. Li, and Y. Wu, "Detecting coordinated bots on twitter using unsupervised graph clustering," *Information Processing & Management*, vol. 59, no. 2, p. 102857, 2022.
- [3] L. Zhang and X. Wu, "Color image interpolation using directional filtering and a posteriori decision," *IEEE Transactions on Image Processing*, vol. 14, no. 10, pp. 1721–1732, 2006.
- [4] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 38, no. 2, pp. 295–307, 2015.
- [5] B. Lim, S. Son, H. Kim, S. Nah, and K. Mu Lee, "Enhanced deep residual networks for single image super-resolution," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 136–144, 2017.
- [6] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 286–301, 2018.
- [7] R. Zhang, P. Isola, and A. A. Efros, "Colorful image colorization," in *European Conference on Computer Vision (ECCV)*, pp. 649–666, Springer, 2016.
- [8] S. Iizuka, E. Simo-Serra, and H. Ishikawa, "Let there be color!: Joint end-to-end learning of global and local image priors," *ACM Transactions on Graphics (TOG)*, vol. 35, no. 4, pp. 1–11, 2016.
- [9] G. Larsson, M. Maire, and G. Shakhnarovich, "Learning representations for automatic colorization," in *European Conference on Computer Vision (ECCV)*, pp. 577–593, Springer, 2016.
- [10] Z. Cheng, Q. Yang, and B. Sheng, "Deep colorization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 415–423, 2015.
- [11] G. Vitoria, A. Romero, and C. Ballester, "Chromagan: Adversarial picture colorization with semantic class distribution," *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 2445–2454, 2020.
- [12] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1646–1654, 2016.
- [13] Y. Mei, Y. Fan, Y. Zhou, Y. Huang, and D. Xu, "Image super-resolution with cross-scale non-local attention and exhaustive self-exemplars mining," *IEEE Transactions on Image Processing*, vol. 29, pp. 1789–1802, 2020.
- [14] J. W. Lee, S. C. Lee, and S. Y. Lee, "Reference-based image super-resolution with deep feature matching," *IEEE Transactions on Image Processing*, vol. 29, pp. 5041–5056, 2020.
- [15] L. Mou and X. X. Zhu, "Deep learning-based supervised remote sensing image colorization using multispectral guidance," *Remote Sensing*, vol. 13, no. 12, p. 2387, 2021.
- [16] I. Kokkinos, "Ubertnet: Training a universal convolutional neural network for low-, mid-, and high-level vision using diverse datasets and limited memory," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6129–6138, 2017.
- [17] S. Vandenheide, S. Georgoulis, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3879–3899, 2021.
- [18] A. R. Zamir, A. Sax, W. Shen, L. J. Guibas, J. Malik, and S. Savarese, "A robust learning approach to multi-task learning," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 18519–18530, 2020.
- [19] Y. Jo, S. Wug Oh, J. Kang, and S. J. Kim, "Deep video super-resolution network using dynamic upsampling filters without explicit motion compensation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3224–3232, 2018.

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